

ECI 39: Exploiting and securing communication on the Internet Day #3

Exploiting and Protecting DNS and SMTP

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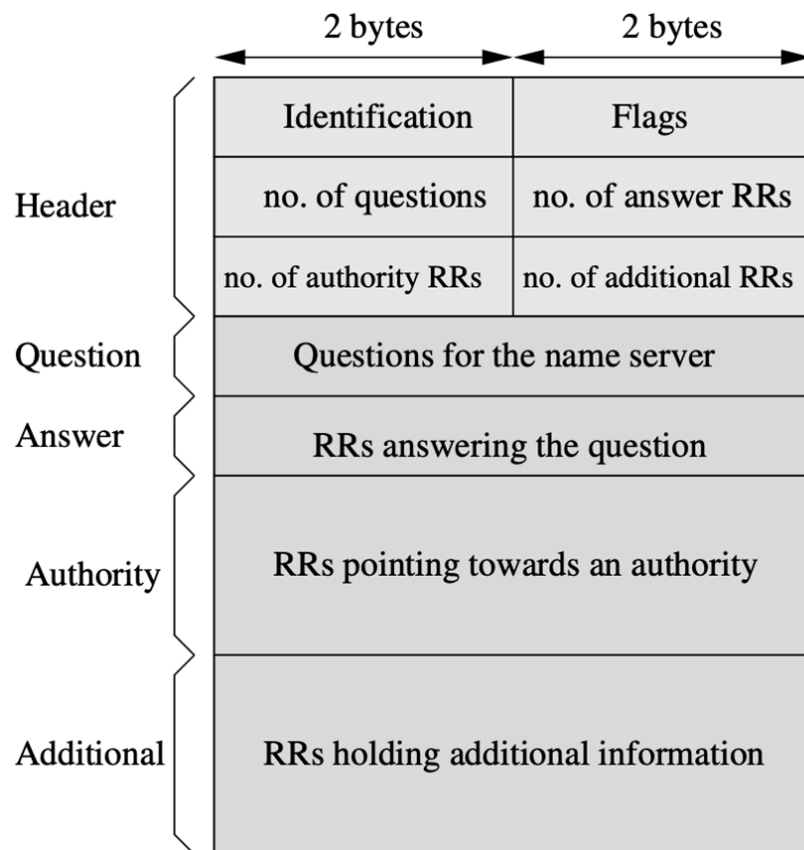
29th July 2026

DOMAIN NAME SYSTEM 101

- DNS is global directory of host names and other services
- It provides a system for translation of domain names into IP addresses
- **DNS architecture**
 - Namespace of DNS addresses + mapping of domain names to IP addresses
 - Storage of DNS namespace
 - Data access and lookups
- The domain names are organized in an inverted searching tree

- Domain names directory is physically stored on DNS servers
 - DNS servers contains information about tree structure
 - Each server manages only portion of domain names - a zone
- Type of servers:
 - **Primary (master, primary name servers)**
 - Contains full authoritative records about managed domains
 - Each domain only one primary server
 - **Secondary (slave, secondary name servers)**
 - Stores authoritative copies from primary servers
 - **Backup (caching-only name servers)**
 - Accept requests and forward to other servers
 - Cache responses
 - Offers non-authoritative responses (can be not actual)

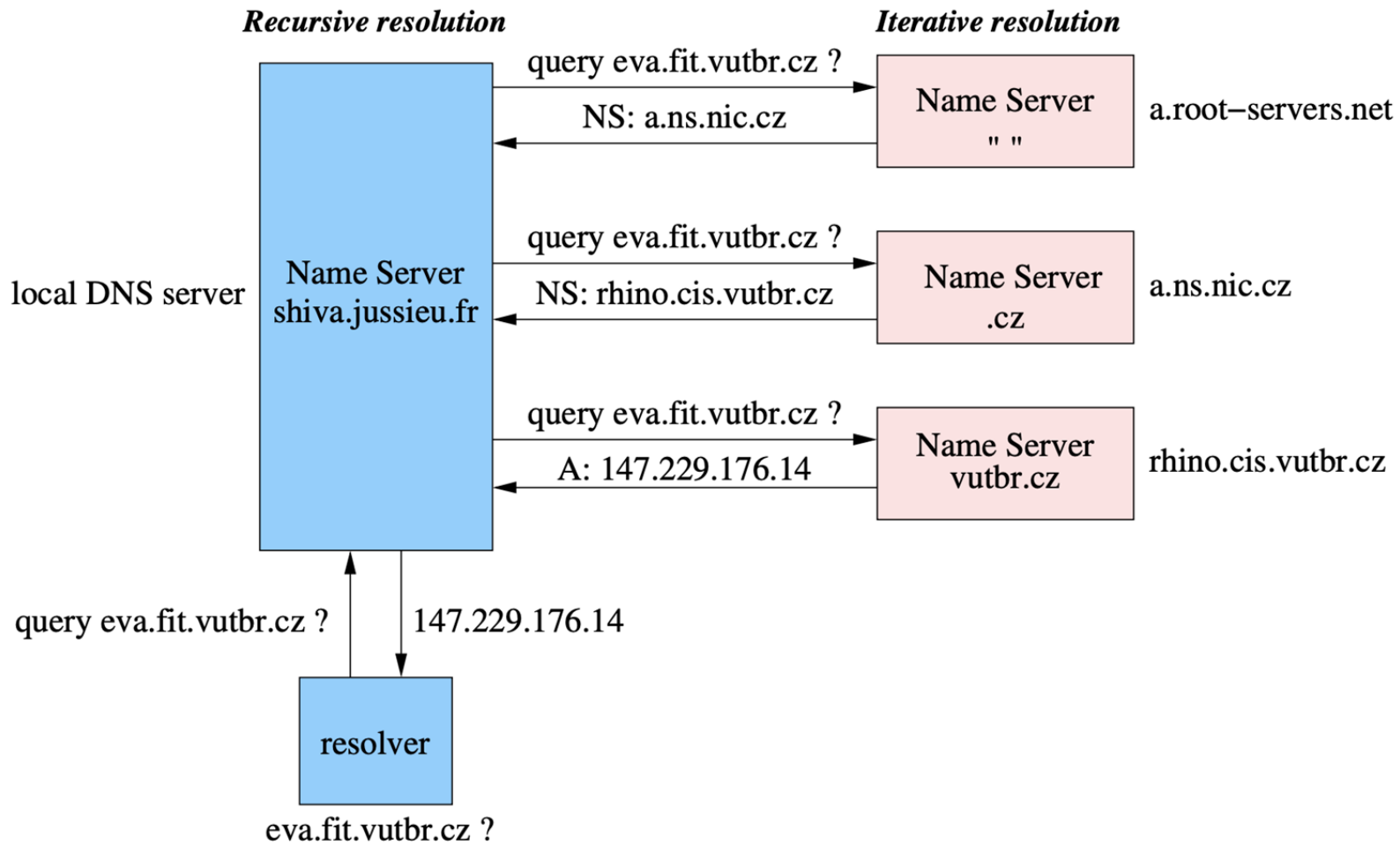
- Introduced in RFC 1035
- Both underlying transport protocols UDP and TCP
 - Primarily UDP used
 - Port 53



Example: answer to www.fit.vutbr.cz query

59058	qr, rd, ra, ad
1	1
4	6
www.fit.vutbr.cz IN A	
www.fit.vutbr.cz 13963 IN A 147.229.9.23	
fit.vutbr.cz. 13869 IN NS guta.fit.vutbr.cz. fit.vutbr.cz. 13869 IN NS kazi.fit.vutbr.cz fit.vutbr.cz 13869 IN NS rhino.cis.vutbr.cz fit.vutbr.cz 13869 IN NS gate.feec.vutbr.cz	
guta.fit.vutbr.cz. 14062 IN A 147.229.8.11 kazi.fit.vutbr.cz. 13869 IN A 147.229.8.12 rhino.cis.futbr.cz. 86211 IN A 147.229.3.10 guta.fit.vutbr.cz. 14156 IN AAAA 2001:67c:1220:809 rhino.cis.vutbr.cz. 43692 IN AAAA 2001:67c:1220:e000:	

- Searching records in DNS
- **DNS Resolver**
 - Communication is of type client - server
 - System routine (resolver) resolves domain names for an application
- **DNS Resolution**
 - Process of searching answer for DNS query in DNS system
 - **Recursive Query**
 - If server do not know the answer, it sends the query to other servers
 - **Iterative Query**
 - If server do not know the answer, responds with a link to a server that can have the answer



- DNS Resource Records format
- Defined in RFC 1034, RFC 1035

Resource Records Format

Name (variable length)
Type (16 bits)
Class (16 bits)
TTL (32 bits)
RDLENGTH (16 bits)
RDATA (variable length)

Example

www.fit.vutbr.cz
CNAME
IN (0x0001)
4106 (1 h 8 min 26 s)
4
tereza.fit.vutbr.cz

- **A (Address)**
 - Direct mapping of domain name to IPv4 address
- **AAAA**
 - Direct mapping of domain name to IPv6 address
- **SOA (Start Of Authority)**
 - Every zone has only one SOA
 - Contains name of primary server, administrators address, several timers (Refresh, Retry, Expire, Minimum) and Serial identifying change of record
- **NS (Name Server)**
 - Determines authoritative server(s) for a zone
- **CNAME (Canonical Name)**
 - Maps alias to canonical name of a host
 - Alias is e.g. www, email, ...

- **PTR (Domain Name Pointer)**
 - Maps IPv4 and IPv6 address to domain name (reverse mapping)
- **NAPTR (Naming Authority Pointer)**
 - Dynamic configuration
 - Uses regular expressions for dynamic records
- **SRV (Service Record)**
 - Localization of services and servers, XMPP, SIP, ...
- **MX (Mail Exchanger)**
 - Contains mail servers to use for a domain
 - Multiple servers possible
- **TXT**
 - Extends the information about domain, server, administrator, ...

whois <domain_name|IP>

- Get the information about the owner of domain or IP address

host|dig <domain_name>

- Do the resolving for given domain name

nslookup [OPTIONS] <domain_name>

- OPTIONS
 - **-type=<record_typeShortcut>**
- example: **nslookup -type=AAAA cisco.com**

- `nslookup uba.ar 8.8.8.8`

Server: dns.google

Address: 8.8.8.8

Non-authoritative answer:

Name: uba.ar

Address: 157.92.5.15

- `nslookup -type=MX fit.vut.cz 1.1.1.1`

Server: one.one.one.one

Address: 1.1.1.1

Non-authoritative answer:

fit.vut.cz MX preference = 10, mail exchanger = adela.fit.vut.cz

fit.vut.cz MX preference = 20, mail exchanger = fiona.fit.vut.cz

- All resolutions are stored in (local) caches

```
ipconfig /displaydns
```

Windows IP Configuration

```
uba.ar
```

```
-----
```

```
Record Name . . . . . : uba.ar
```

```
Record Type . . . . . : 1
```

```
Time To Live . . . . . : 86396
```

```
Data Length . . . . . : 4
```

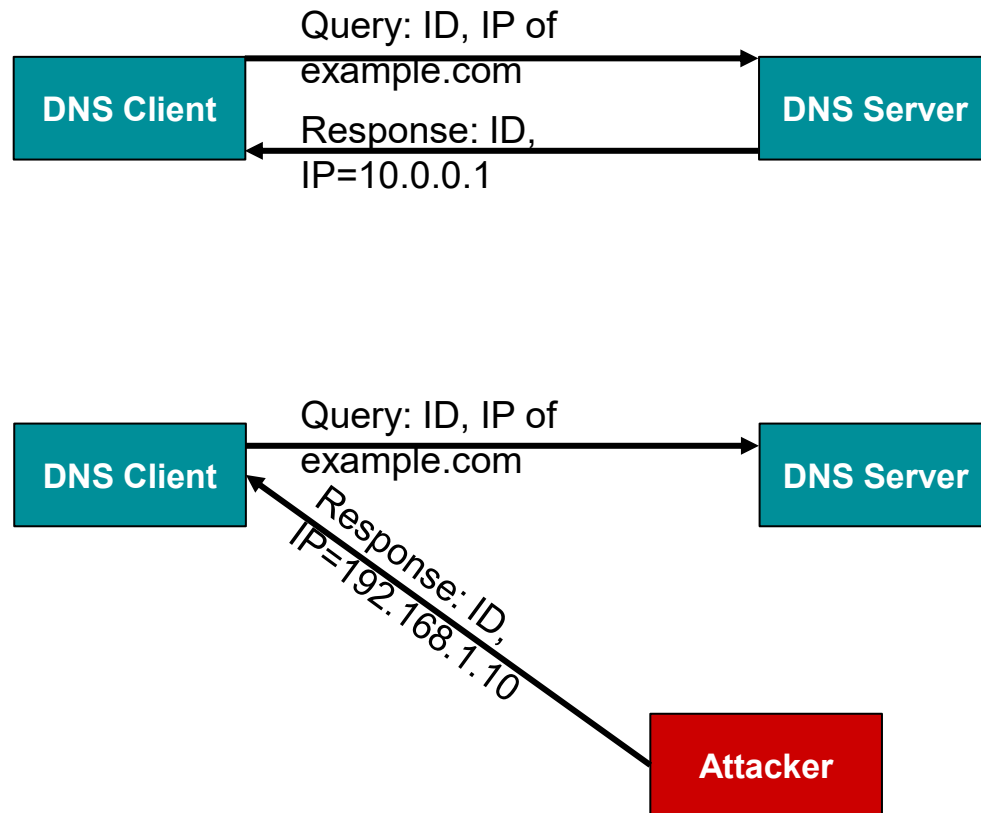
```
Section . . . . . : Answer
```

```
A (Host) Record . . . : 157.92.5.15
```

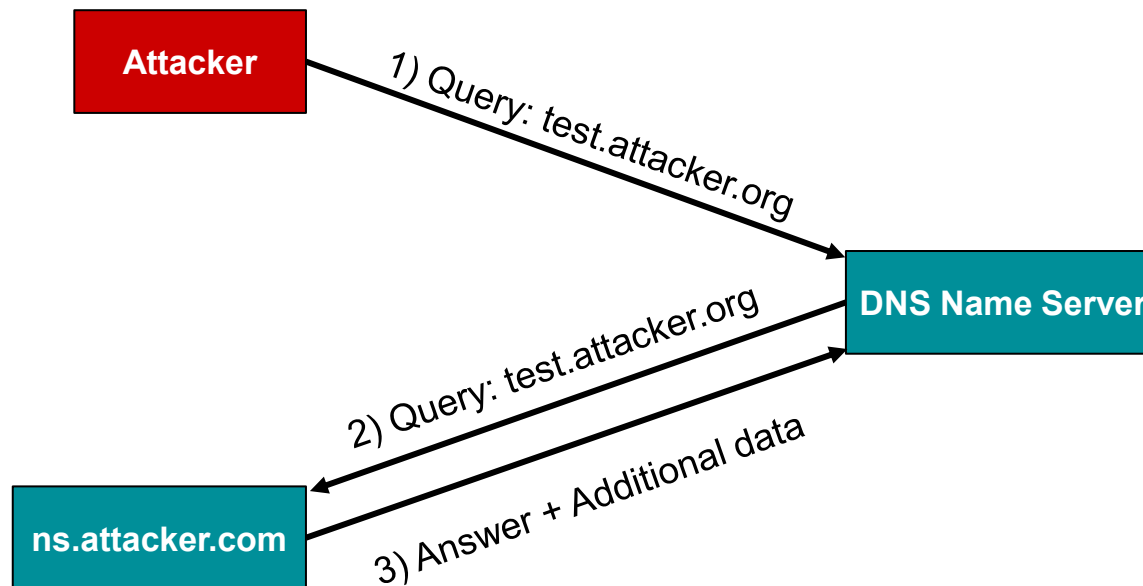
- Distributed Denial of Service (DDoS)
- DDoS against root servers
- Blocking service for legitimate responses
- <https://root-servers.org/>



- Unauthorized response for client query from the attacker



- DNS cache poisoning
- Insertion of incorrect information into cache



- DNS does not encrypt the traffic
- So anyone who can sniff the traffic can check the content of DNS packets
- This effect is used by network administrators to manage users
 - Point to right servers (in companies)
 - Or to filter the traffic
- But ISPs can also monitor (spy) the traffic
- Privacy issue


ADGUARD HOME ON

[Dashboard](#)
[Settings](#)
[Filters](#)
[Query Log](#)
[Setup Guide](#)
[Sign out](#)

Query Log

Filtered

Time	Request	Response	Client
14:43:16 7/29/2026	 www.bing.com Type: A, Plain DNS	 Safe Search 0.79 ms	 10.142.210.3
14:43:13 7/29/2026	 www.google.com Type: A, Plain DNS	 Safe Search 1 ms	 10.142.210.3
14:43:10 7/29/2026	 www.google-analytics.com Type: A, Plain DNS	 Blocked AdGuard DNS filter	 10.142.210.3
14:43:10 7/29/2026	 comcluster.cxense.com Type: A, Plain DNS	 Blocked OISD Blocklist Big	 10.142.210.3
14:42:59 7/29/2026	 spir.hit.gemius.pl Type: A, Plain DNS	 Blocked OISD Blocklist Big	 10.142.210.3
14:42:59 7/29/2026	 gacz.hit.gemius.pl Type: A, Plain DNS	 Blocked OISD Blocklist Big	 10.142.210.3
14:42:59 7/29/2026	 delivery.r2b2.cz Type: A, Plain DNS	 Blocked AdGuard DNS filter	 10.142.210.3
14:42:59 7/29/2026	 cdn.cpex.cz Type: A, Plain DNS	 Blocked OISD Blocklist Big	 10.142.210.3
14:42:59 7/29/2026	 ... Type: A, Plain DNS	 Blocked by CNAME or IP	 10.142.210.3

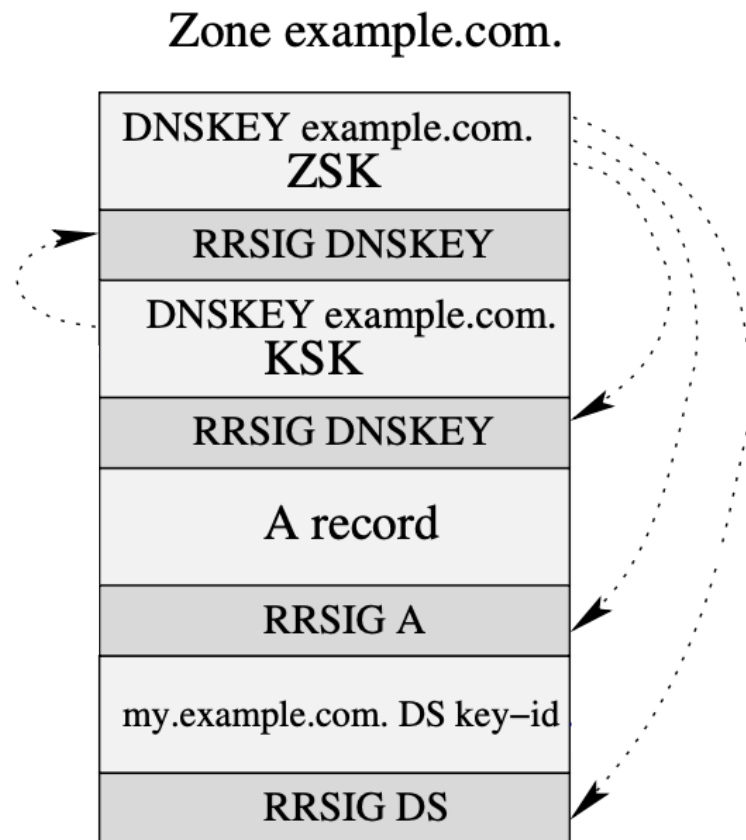
Fixing vulnerabilities 2 and 3

SIGNING DNS RECORDS

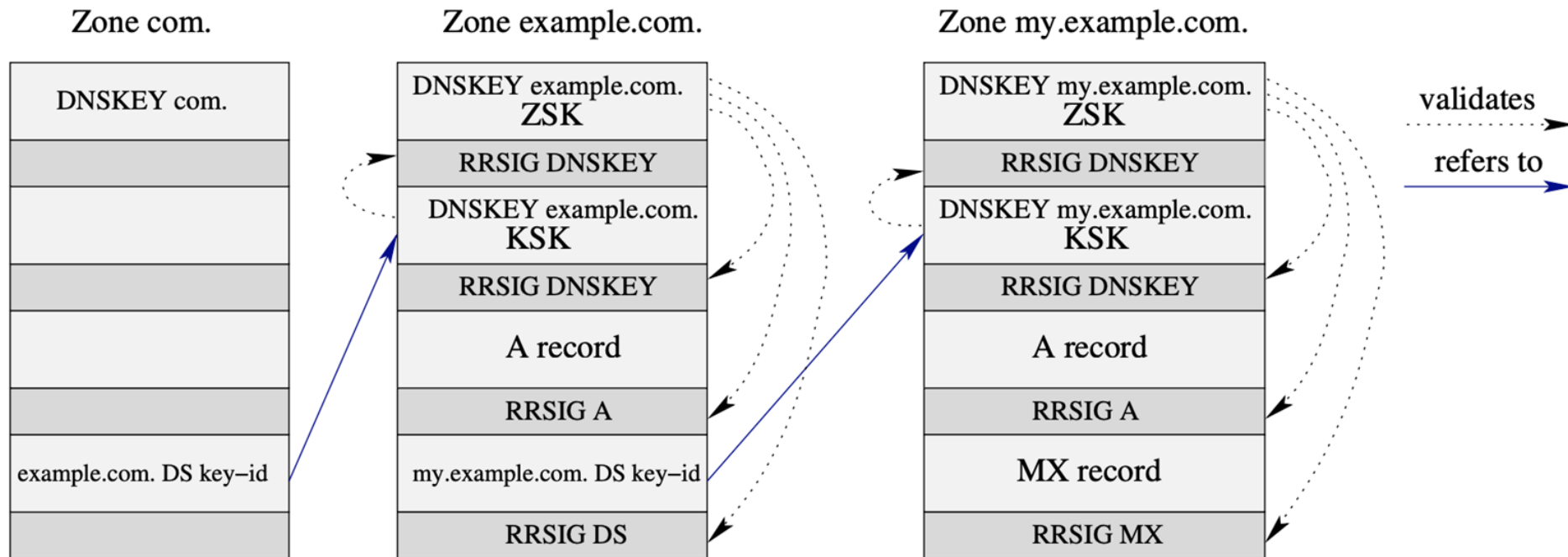
- Covered by RFC 4033, RFC 4034, RFC 4035, RFC 5155
- Adds integrity and authentication to DNS data
- Asymmetric cryptography used for DNS record signing
- **Zone Signing Key (ZSK)**
 - Public, private key
 - Used for signing DNS records in zones
- **Key Signing Key (KSK)**
 - Public, private key
 - Used for signing keys that are used for signing records

- New records are defined for DNSSEC
- **DNSKEY**
 - Public key for signature verification
- **RRSIG**
 - Signature of the record
- **NSEC, NSEC3**
 - Link to another record when the queried domain does not exists
- **DS**
 - Record for verification of DNSKEY record
 - Stored in superior domain

- **ZSK**
 - **Public Key**
 - Stored in DNSKEY record used to validate signatures in RRSIGs
 - **Private Key**
 - Used to sign zone records signatures stored in RRSIGs
- **KSK**
 - **Public Key**
 - Stored in DS record used to validate DNSKEY
 - **Private Key**
 - Used to sign DNSKEY

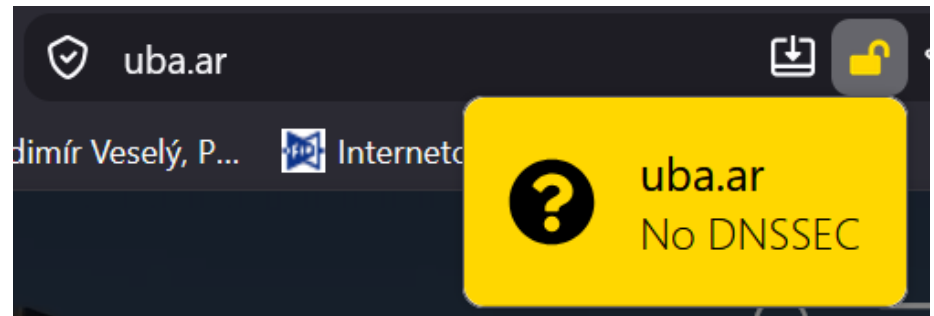
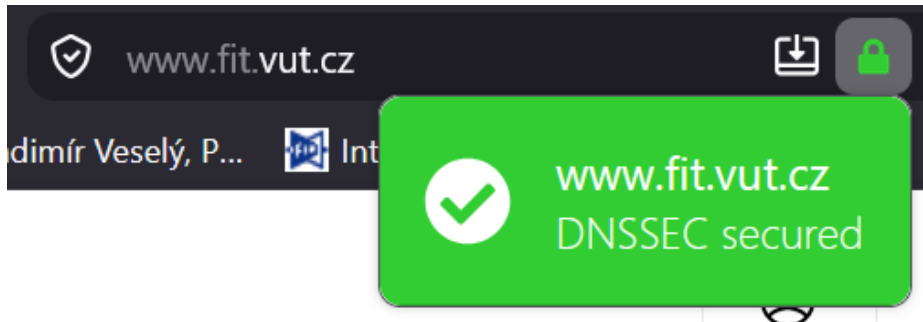


- Chain of trust
- Verification of key authenticity by the superior domain
- Public and Private key pair exists for every zone
- DNSSEC records are significantly increasing packet payload size



- **DNSSEC does not provide encryption!**
 - Everyone who will sniff the traffic can read the content of DNS messages
 - DNSSEC provides only integrity and authentication but not encryption
 - *It does not solves privacy issues!*
- *What can we use to encrypt DNS traffic?*
 - There are few attempts to encapsulate DNS messages to other encrypted protocols
 - DNS over TLS
 - DNS over HTTPS
 - DNS over QUIC

- Various add-ons and plug-ins
 - Unfortunately, nothing on OS level 😞



Fixing vulnerability 4

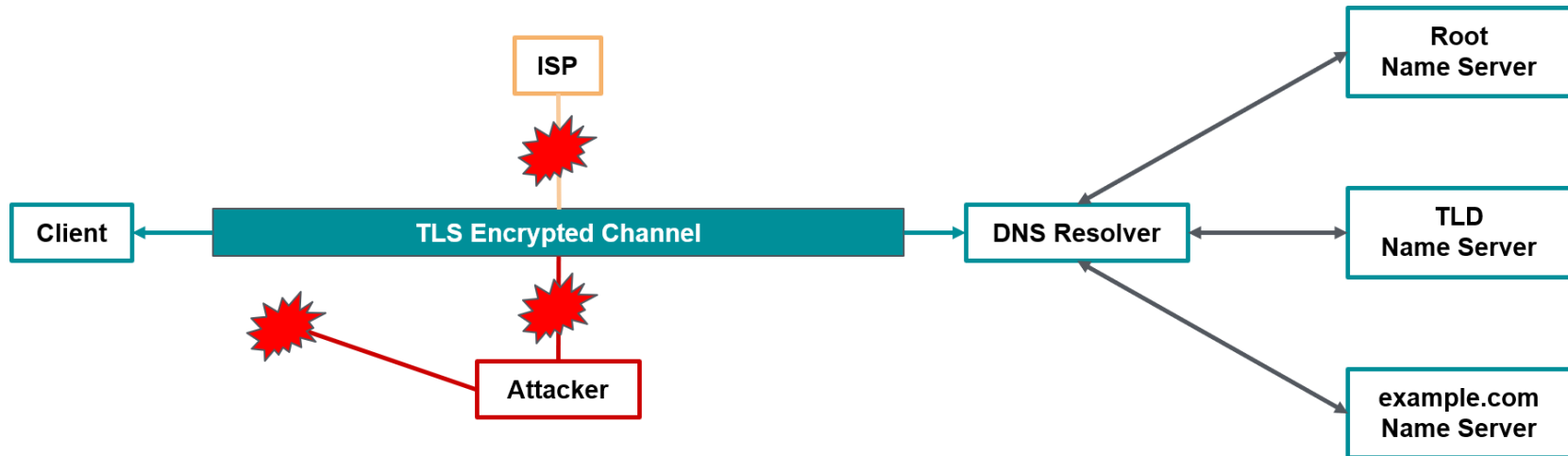
SECURING DNS TRAFFIC

- Standard DNS queries and responses as defined in DNS are encapsulated by TLS protocol
- TLS protocol is used for encryption and transport of DNS messages
- Multiple queries can be transmitted over the same encrypted channel



- Defined in RFC 7858
- **Dedicated port 853/tcp**
- Support on the OS level, proxies and resolvers

- DoT moves the missing privacy to other party
- DNS traffic is encrypted only between Client (browser) and the DNS Resolver



- *Can we trust the DNS Resolver provider? Is it really private?*
- *Are those connections between DNS Resolver and Name Servers secure?*

- kdig

```
$ kdig -d @1.1.1.1 +tls-ca +tls-host=cloudflare-dns.com example.com
;; DEBUG: Querying for owner(example.com.), class(1), type(1), server(1.1.1.1), port(853), protocol(TCP)
;; DEBUG: TLS, imported 170 system certificates
;; DEBUG: TLS, received certificate hierarchy:
;; DEBUG:  #1, C=US,ST=CA,L=San Francisco,O=Cloudflare\, Inc.,CN=*.cloudflare-dns.com
;; DEBUG:      SHA-256 PIN: yioEpqeR4WtDwE9YxNVnCEkTxIjx6EEIwFSQW+lJsbc=
;; DEBUG:  #2, C=US,O=DigiCert Inc,CN=DigiCert ECC Secure Server CA
;; DEBUG:      SHA-256 PIN: PZXN3lRAy+8tBKk20x6F7jIlzr2Yzmqc3JnyfXoCw=
;; DEBUG: TLS, skipping certificate PIN check
;; DEBUG: TLS, The certificate is trusted.
;; TLS session (TLS1.3)-(ECDHE-SECP256R1)-(ECDSA-SECP256R1-SHA256)-(AES-256-GC
;; ->>HEADER<<- opcode: QUERY; status: NOERROR; id: 58548
;; Flags: qr rd ra; QUERY: 1; ANSWER: 1; AUTHORITY: 0; ADDITIONAL: 1

;; EDNS PSEUDOSECTION:
;; Version: 0; flags: ; UDP size: 1536 B; ext-rcode: NOERROR
;; PADDING: 408 B

;; QUESTION SECTION:
;; example.com.                IN  A

;; ANSWER SECTION:
example.com.                2347    IN  A    93.184.216.34

;; Received 468 B
;; Time 2018-03-31 15:20:57 PDT
;; From 1.1.1.1@853(TCP) in 12.6 ms
```

- <https://developers.cloudflare.com/1.1.1.1/dns-over-tls/>

- Use TLS to encrypt the traffic
- Encapsulates DNS request/response by the HTTP
 - => HTTPS
- **Uses TCP transport port 443!**
 - The same port as for other HTTPS



- Defined in RFC 8484
- Brings encryption to DNS
- Can be enabled on the application level (browsers, proxies, tools)
- Currently supported in almost all browsers and mobile OS
- Hard to detect
 - Port based identification impossible
 - Employment of ML methods, blocklists, active scanning
- RFC recommended minimum HTTP/2
 - Adds additional establishment overhead

- HTTP accept type: "application/dns-message"
- Payload: DNS query in DNS Wireformat

```
:method = POST
:scheme = https
:authority = dnsserver.example.net
:path = /dns-query
accept = application/dns-message
content-type = application/dns-message
content-length = 33
```

```
<33 bytes represented by the following hex encoding>
00 00 01 00 00 01 00 00 00 00 00 00 03 77 77 77
07 65 78 61 6d 70 6c 65 03 63 6f 6d 00 00 01 00
01
```

- <https://tools.ietf.org/html/rfc8484>

- HTTP accept type: "application/dns-message"
- Query encoded using base64url without padding
 - Format: /dns-query?dns=<base64url_encoded>

```
:method = GET
:scheme = https
:authority = dnsserver.example.net
:path = /dns-query? (no space or Carriage Return (CR))
      dns=AAABAAABAAAAAAAAAWE-NjJjaGFyYWN0ZXJsYWJl (no space or CR)
      bC1tYWtlcyliYXNlNjRlcmwtZGlzdGluY3QtZnJvbS1z (no space or CR)
      dGFuZGFyZC1iYXNlNjQHZXhxbXBsZQNjb20AAAEAAQ
accept = application/dns-message
```

- Standardised (RFC 8484)
- HTTP accept type: "application/dns-message"
- Success 2xx with query response
- Unsuccessful response 4xx without any query response

```
:status = 200
content-type = application/dns-message
content-length = 61
cache-control = max-age=3709
```

<61 bytes represented by the following hex encoding>

```
00 00 81 80 00 01 00 01 00 00 00 00 03 77 77 77
07 65 78 61 6d 70 6c 65 03 63 6f 6d 00 00 1c 00
01 c0 0c 00 1c 00 01 00 00 0e 7d 00 10 20 01 0d
b8 ab cd 00 12 00 01 00 02 00 03 00 04
```

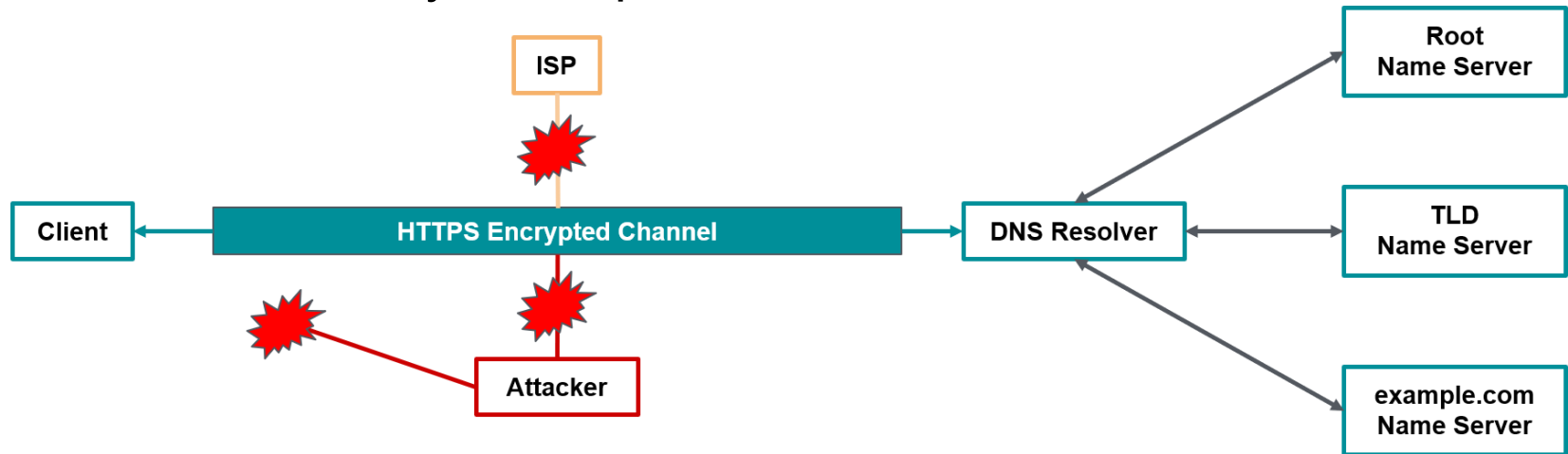
- **Not standardised**
- Cloudflare, Google - similar request, the same response
- Http accept type "application/dns-json"

```
curl -H 'accept: application/dns-json' 'https://cloudflare-dns.com/dns-query?name=example.com&type=AAAA'
```

```
{
  "Status": 0,
  "TC": false,
  "RD": true,
  "RA": true,
  "AD": true,
  "CD": false,
  "Question": [
    {
      "name": "example.com.",
      "type": 28
    }
  ],
  "Answer": [
    {
      "name": "example.com.",
      "type": 28,
      "TTL": 1726,
      "data": "2606:2800:220:1:248:1893:25c8:1946"
    }
  ]
}
```

- <https://developers.cloudflare.com/1.1.1.1/dns-over-https/json-format/>

- DoH provides similar functionality as DoT
- DoH has higher payload overhead, longer initiation phase
- Does it solve any of the problems of DoT?

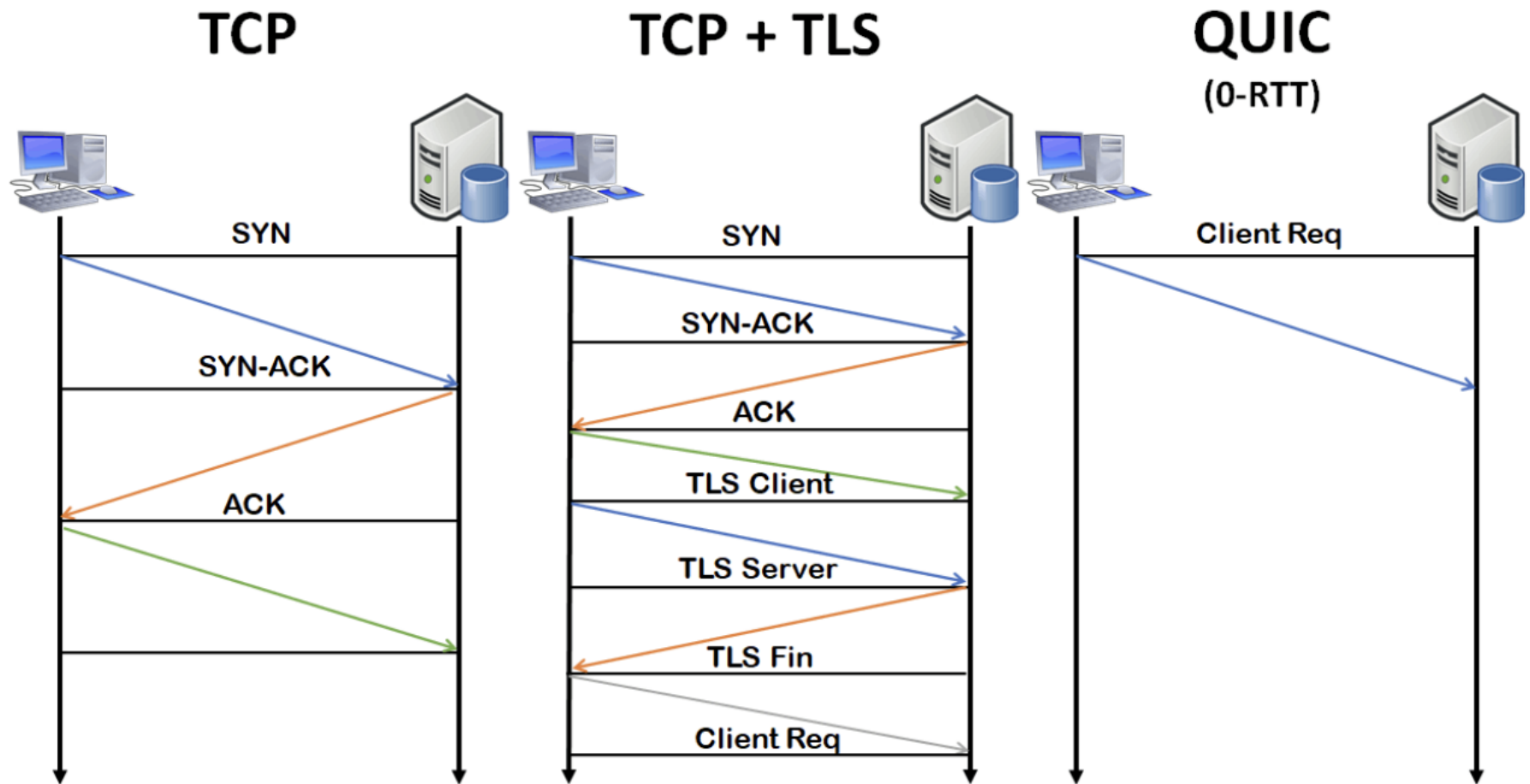


- Similar privacy concerns remain:
 - *Can we trust the DNS Resolver provider? Is it really private?*
 - *Are those connections between DNS Resolver and Name Servers secure?*

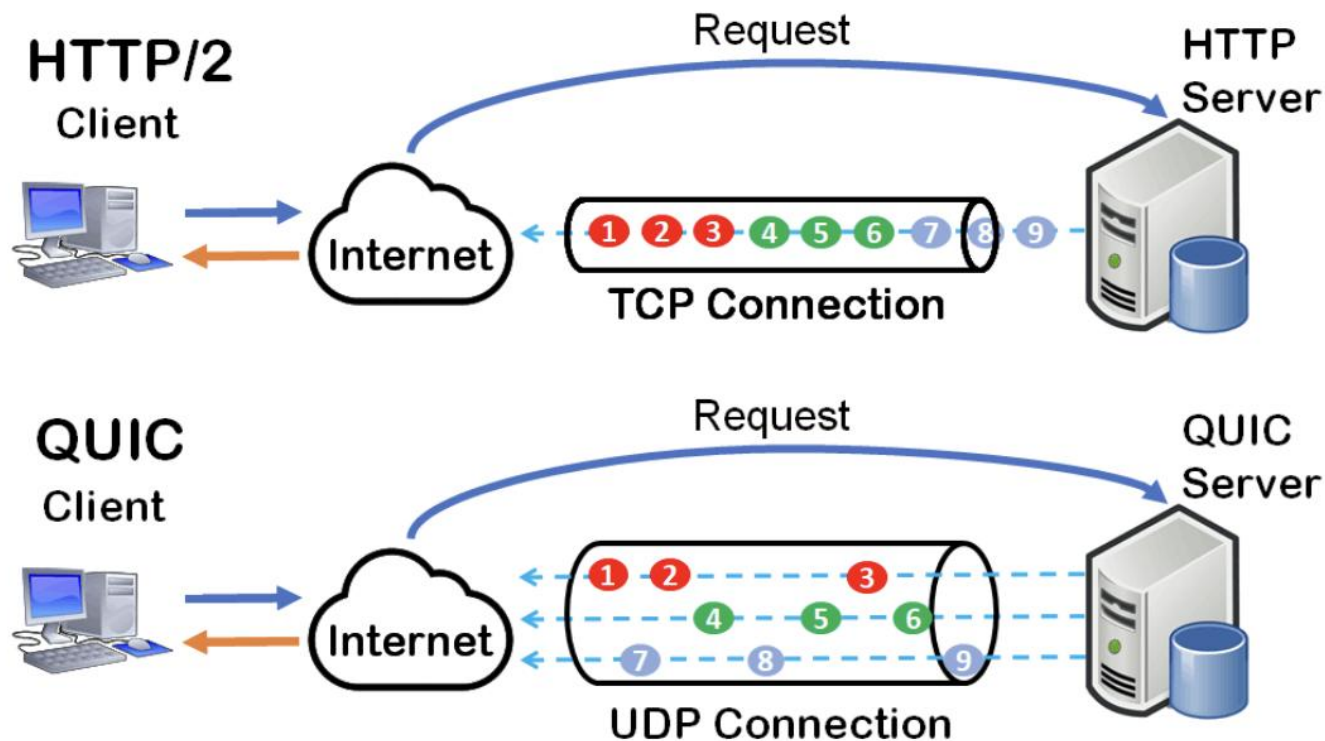
```
netsearch@POCITAC /m/c/U/Kvetak> curl -s -H 'accept: application/dns-json'
'https://cloudflare-dns.com/dns-query?name=uba.ar&type=A' | jq
{
  "Status": 0,
  "TC": false,
  "RD": true,
  "RA": true,
  "AD": false,
  "CD": false,
  "Question": [
    {
      "name": "uba.ar",
      "type": 1
    }
  ],
  "Answer": [
    {
      "name": "uba.ar",
      "type": 1,
      "TTL": 86400,
      "data": "157.92.5.15"
    }
  ],
  "Comment": [
    "EDE(18): Prohibited"
  ]
}
```

- Defined in RFC 9250
- Use QUIC to transport DNS
- **Use dedicated port 853/udp**

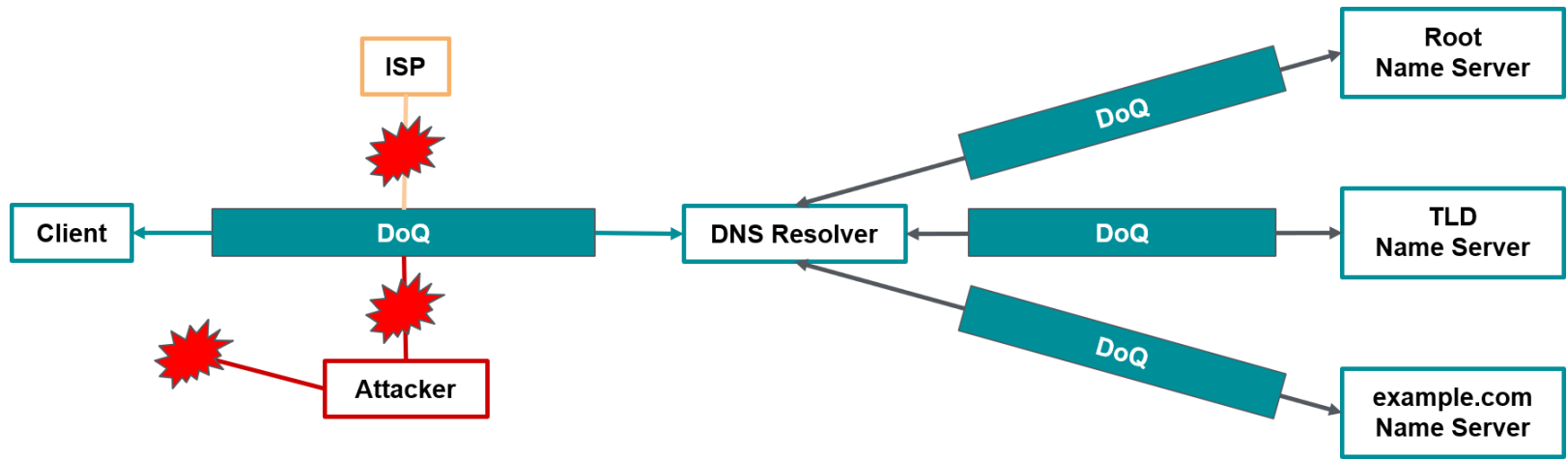




- Ahead-of-line blocking
- <https://avenash-kumar.com/blog/quic-birth-of-new-protocol-in-layer4-family>



- DoQ provides similar functionality as DoT but even more
- DoQ has less payload overhead than DoH
- Does it solve any of the problems of previous mentioned?



- Similar privacy concerns remains:
 - *Can we trust the DNS Resolver provider? Is it really private?*
 - *Are those connections between DNS Resolver and Name Servers secure?*

< DNS over HTTPS

Advanced settings

Domain Name System over HTTPS (DoH) encrypts site lookups so it's harder for your internet provider or others to see what websites you're about to visit.

[Learn more](#)

✔ DNS over HTTPS is using the provider Cloudflare

☐ Default

Use secure DNS in regions where it's available

☒ Custom

Always use secure DNS with control over your provider and fallback behavior

☒ Always warn me if secure DNS isn't available

Choose provider:

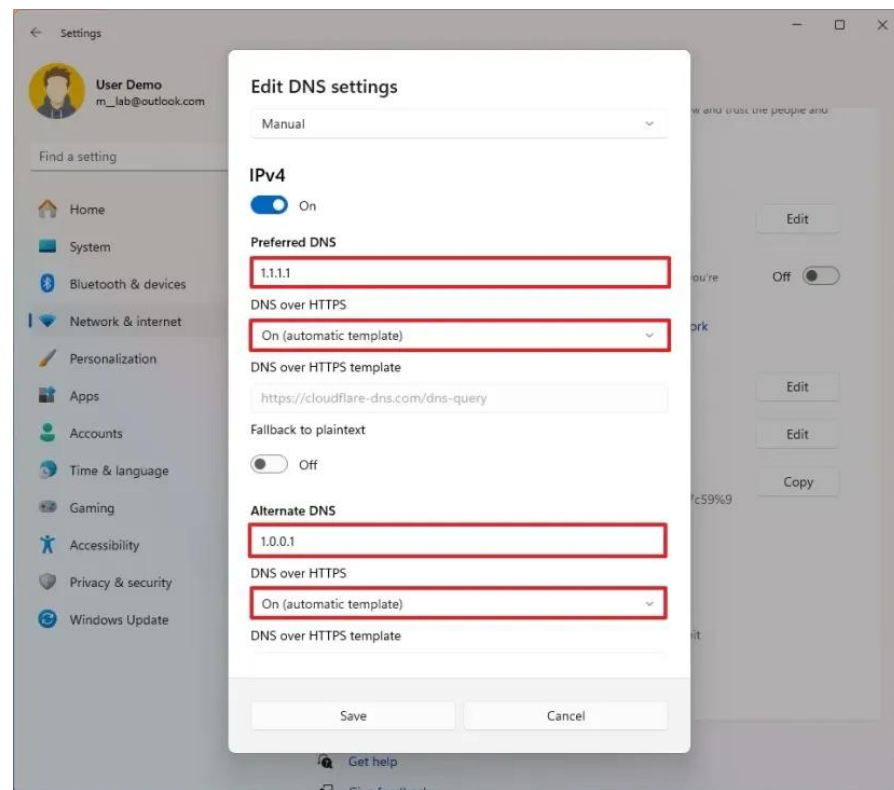
Custom

`https://mozilla.cloudflare-dns.com/dns-query`

☐ Off

Use your default DNS resolver

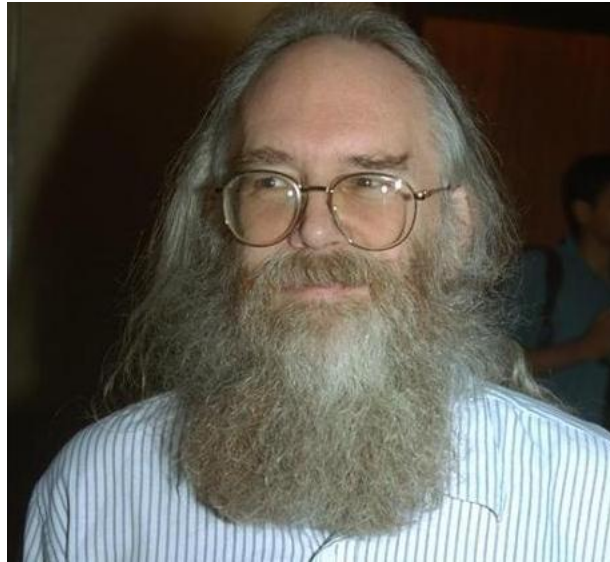
Manage exceptions



EMAIL 101

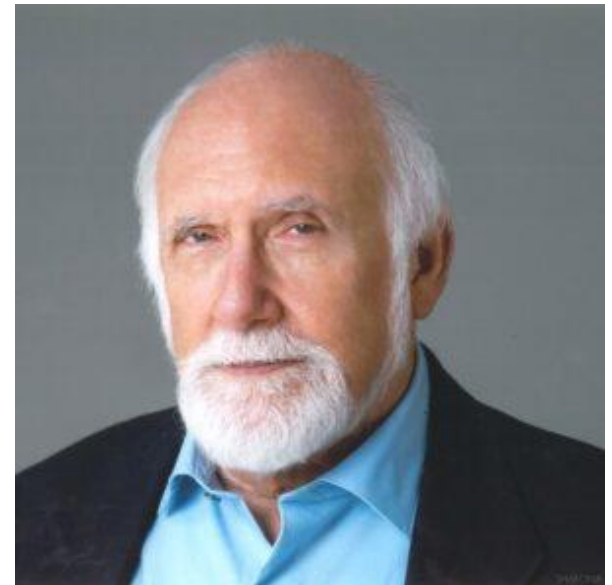
- An estimated **392.5 billion emails** are sent and received every day worldwide in **2026**, up from 376.4 billion in 2025 — a similar ~4.3% year-over-year increase. There are approximately 4.7 billion active email users in 2026, **about 58% of the global population** There are over **8.3 billion active email accounts worldwide** (Radicati Group, 2025)
- Seznam.cz has 25 million accounts (2022, unchanged — no newer figure available); Gmail has about 1.8 billion active users worldwide as of late 2024/2025
- Regarding the number of email accounts per individual, the average is now reported at roughly **1.86 email accounts per user** (Radicati Group, 2025)
Percentage of emails opened on mobile: **roughly 60% globally**, though some vendor-specific reports (e.g. Gmail) put mobile opens as high as 75–85%
- On average, an office worker still receives about **121 business emails daily**
 - The average emails sent per day at work by an office employee remains around 40 per day
- [Hostinger](#) and [Mailover](#) and [SQ Magazine](#)

- RFC 821
- RFC 2821
- RFC 5321



- Jon Postel

- John Klensin

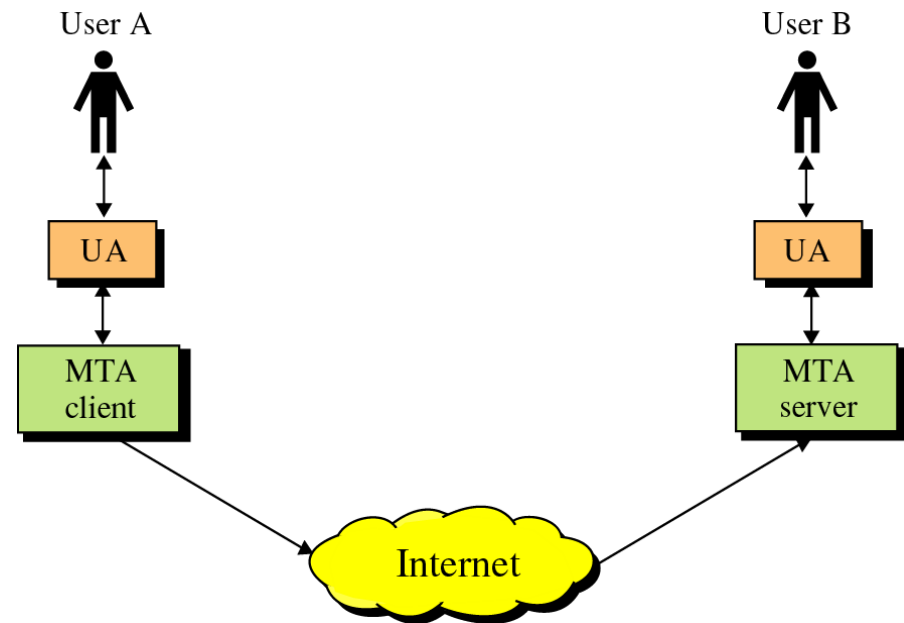


- **User Agent**

- An application with user interface used by individuals to compose, send, receive, and manage email messages.
- It typically interacts with mail servers via protocols like SMTP (for sending) and IMAP or POP (for receiving).

- **MTA**

- Software system responsible for transferring email messages between servers using standardized protocol.
- MTAs work behind the scenes to route and deliver emails from the sender's server to the recipient's server, handling tasks like queue management, retry mechanisms, and error reporting

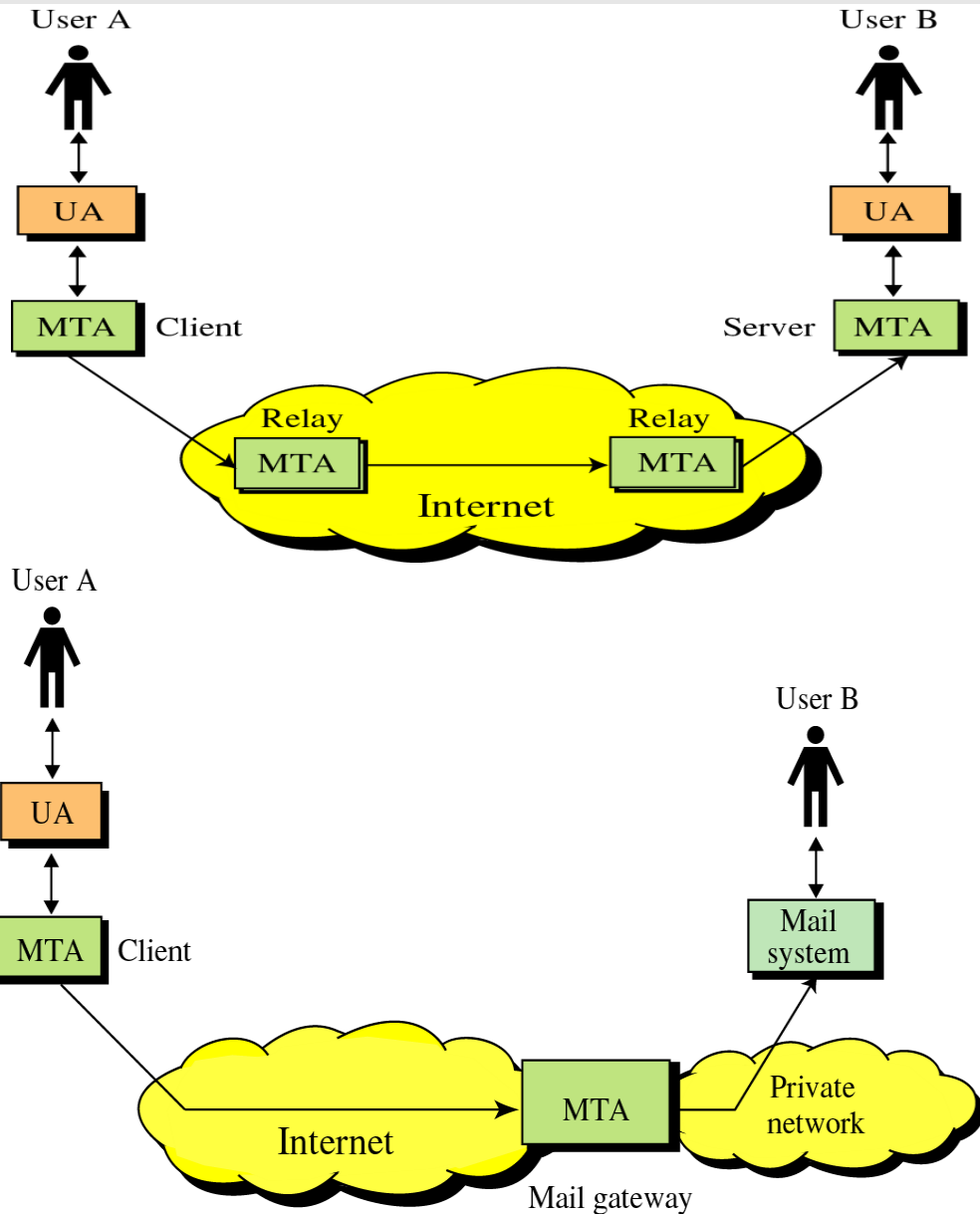


- **Relays**

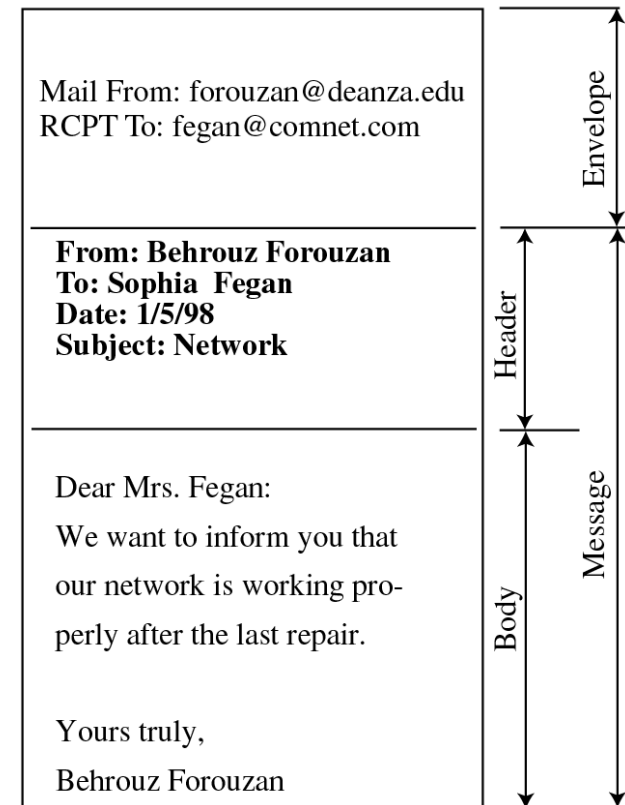
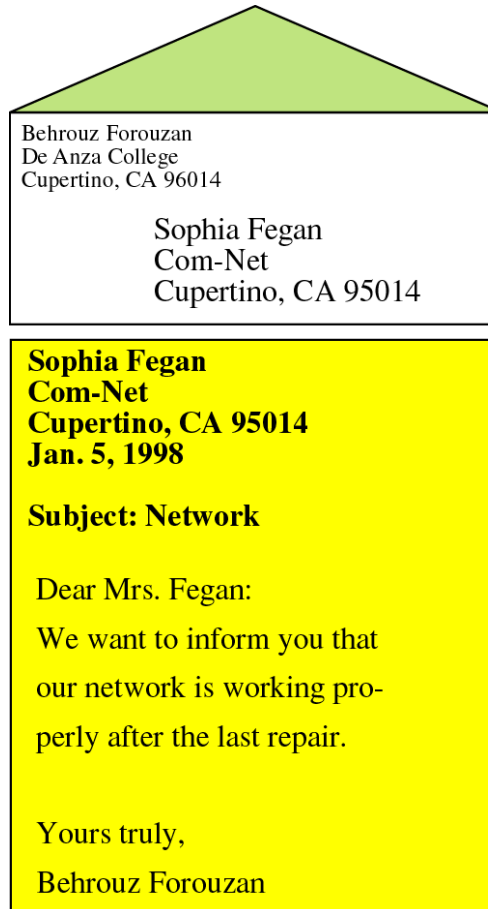
- Mail server that passes email messages from one server to another.
- It acts as an intermediary in the email delivery process, forwarding messages to their final destination.
- Often used by organizations to route outbound emails, mail relays must be carefully configured to prevent abuse (such as spamming) by unauthorized users.

- **Gateway**

- Server or system that serves as an entry or exit point for email traffic, performing various functions such as filtering, security scanning, and routing between different email systems or networks.
- Essentially, it acts as a bridge between internal and external email communications, helping to enforce security policies, detect spam or malware, and ensure proper email delivery.



- Mail is a file
- **Envelope**
 - sender address
 - receiver address
 - other information
- **Message**
 - Mail Header – defines the sender, the receiver, the subject of the message, and some other information
 - Mail Body – Contains the actual information in the message

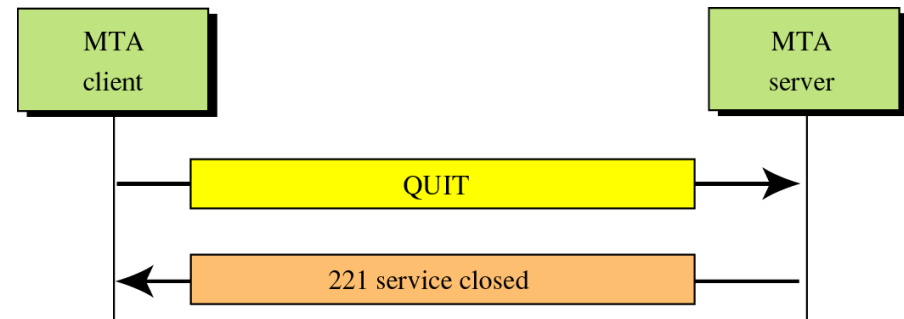
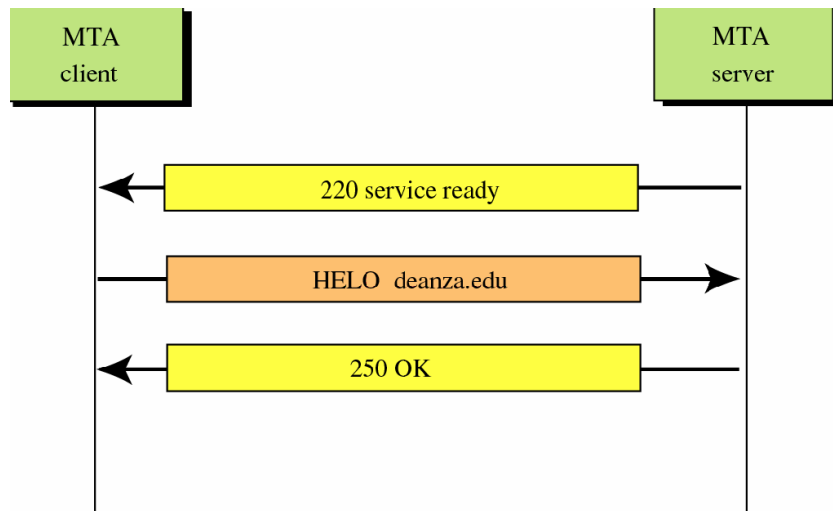


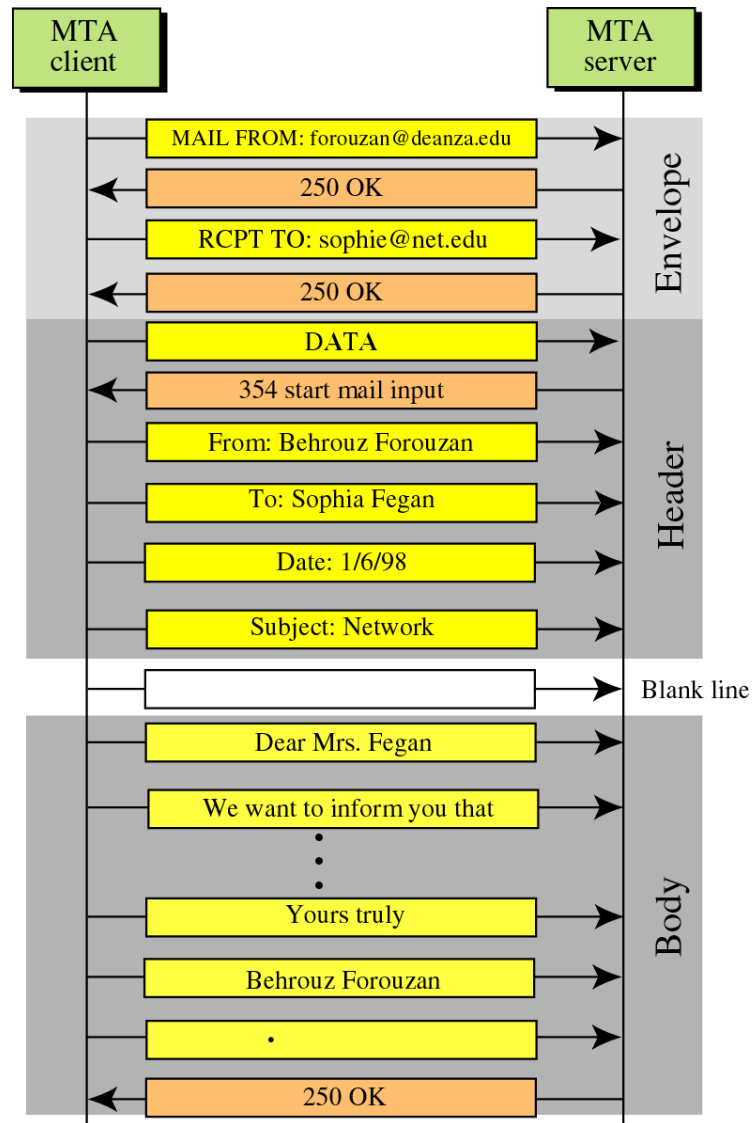
- **MIME** is a standard that extends the format of email to support text in character sets beyond ASCII, as well as attachments of multimedia content like images, audio, video, and application files.
- **Content-Type**
 - `text/plain,`
`text/html,`
`image/jpeg,`
`application/pdf`
- **Encoding**
 - `base64`
- **Multi-part Messages**
 - `multipart/mixed,`
`multipart/alternative`

- runs over TCP
- Well-known ports
 - 25
 - 465: TLS
 - 587: STARTTLS
- STARTTLS command
 - upgrade of connection for SMTP, POP, IMAP
- Authentication
 - No auth
 - Insecure password
 - Encrypted password
 - Kerberos
 - NTLM

Keyword	Command form	Description	Usage
HELO	HELO<SP><domain><CRLF>	It provides the identification of the sender i.e. the host name.	Mandatory
MAIL	MAIL<SP>FROM : <reverse-path><CRLF>	It specifies the originator of the mail.	Mandatory
RCPT	RCPT<SP>TO : <forward-path><CRLF>	It specifies the recipient of mail.	Mandatory
DATA	DATA<CRLF>	It specifies the beginning of the mail.	Mandatory
QUIT	QUIT<CRLF>	It closes the TCP connection.	Mandatory
RSET	RSET<CRLF>	It aborts the current mail transaction but the TCP connection remains open.	Highly recommended
VRFY	VRFY<SP><string><CRLF>	It is use to confirm or verify the user name.	Highly recommended
NOOP	NOOP<CRLF>	No operation	Highly recommended
TURN	TURN<CRLF>	It reverses the role of sender and receiver.	Seldom used
EXPN	EXPN<SP><string><CRLF>	It specifies the mailing list to be expanded.	Seldom used
HELP	HELP<SP><string><CRLF>	It send some specific documentation to the system.	Seldom used
SEND	SEND<SP>FROM : <reverse-path><CRLF>	It send mail to the terminal.	Seldom used
SOML	SOML<SP>FROM : <reverse-path><CRLF>	It send mail to the terminal if possible; otherwise to mailbox.	Seldom used
SAML	SAML<SP>FROM : <reverse-path><CRLF>	It send mail to the terminal and mailbox.	Seldom used

Status Code	Message/Category	Description
2xx Success		
211	System status, or system help reply	Provides system status or help information.
214	Help message	A help message explaining the server's commands.
220	Service ready	The server is ready to accept a new connection (e.g., at connection start-up).
221	Service closing transmission channel	The server is closing the connection after a command (e.g., after QUIT).
250	Requested mail action okay, completed	Indicates successful completion of the requested action.
251	User not local; will forward to <forward-path>	The recipient is not local, but the server will forward the message.
252	Cannot VRFY user, but will accept message	The server cannot verify the recipient, but will attempt delivery anyway.
3xx Redirection		
354	Start mail input; end with <CRLF>.<CRLF>	The server is ready to receive the message body. Terminate input with a single period on a line.
4xx Temporary Failure		
421	Service not available, closing transmission channel	The server is temporarily unavailable; connection will be closed.
450	Requested action not taken: mailbox unavailable	The recipient's mailbox is unavailable (e.g., busy or temporarily inaccessible).
451	Requested action aborted: local error in processing	A local error prevented the action from being completed.
452	Requested action not taken: insufficient system storage	The server lacks sufficient storage space to complete the request.
5xx Permanent Failure		
500	Syntax error, command unrecognized	The server did not recognize the command due to a syntax error.
501	Syntax error in parameters or arguments	There is a syntax error in the command's parameters or arguments.
502	Command not implemented	The server does not support the command.
503	Bad sequence of commands	Commands were sent in an improper sequence.
504	Command parameter not implemented	A command parameter is not supported by the server.
550	Requested action not taken: mailbox unavailable	The requested action failed because the recipient's mailbox is unavailable (e.g., nonexistent or restricted).
551	User not local; please try <forward-path>	The recipient is not local; the sender should try another route for delivery.
552	Requested mail action aborted: exceeded storage allocation	The message size exceeds the storage allocation available for the recipient.
553	Requested action not taken: mailbox name not allowed	The email address syntax is incorrect, or the mailbox name is not allowed.
554	Transaction failed	A general error indicating that the transaction has failed, often due to security or policy restrictions.





- **DO NOT DO THAT IN REAL LIFE!!!**

```
veselyv@merlin: ~$ telnet adela.fit.vutbr.cz 25
Trying 2001:67c:1220:808::93e5:80d...
Connected to adela.fit.vutbr.cz.
Escape character is '^]'.
220 adela.fit.vutbr.cz ESMTP Postfix (Debian)
HELO adela.fit.vutbr.cz
250 adela.fit.vutbr.cz
MAIL FROM:<mgreg@fit.vutbr.cz> BODY=7BIT
250 2.1.0 Ok
RCPT TO:<veselyv@fit.vutbr.cz>
250 2.1.5 Ok
DATA
354 End data with <CR><LF>.<CR><LF>
From:<dona@d@president-trump.com>
To:<j.milei@argentina.ar>
Subject: Selling Patagonia?
Hello, Javier!
Is Patagonia for sale? I want my own penguins...
Yours truly, Donald!
.
250 2.0.0 Ok: queued as A67B826046C
QUIT
221 2.0.0 Bye
Connection closed by foreign host.
```

- RFC 3501
- RFC 9501
- Version 4rev2

- Mark Crispin



- RFC 4550
 - LEMONADE profile
 - P-IMAP



- Alexey Melnikov

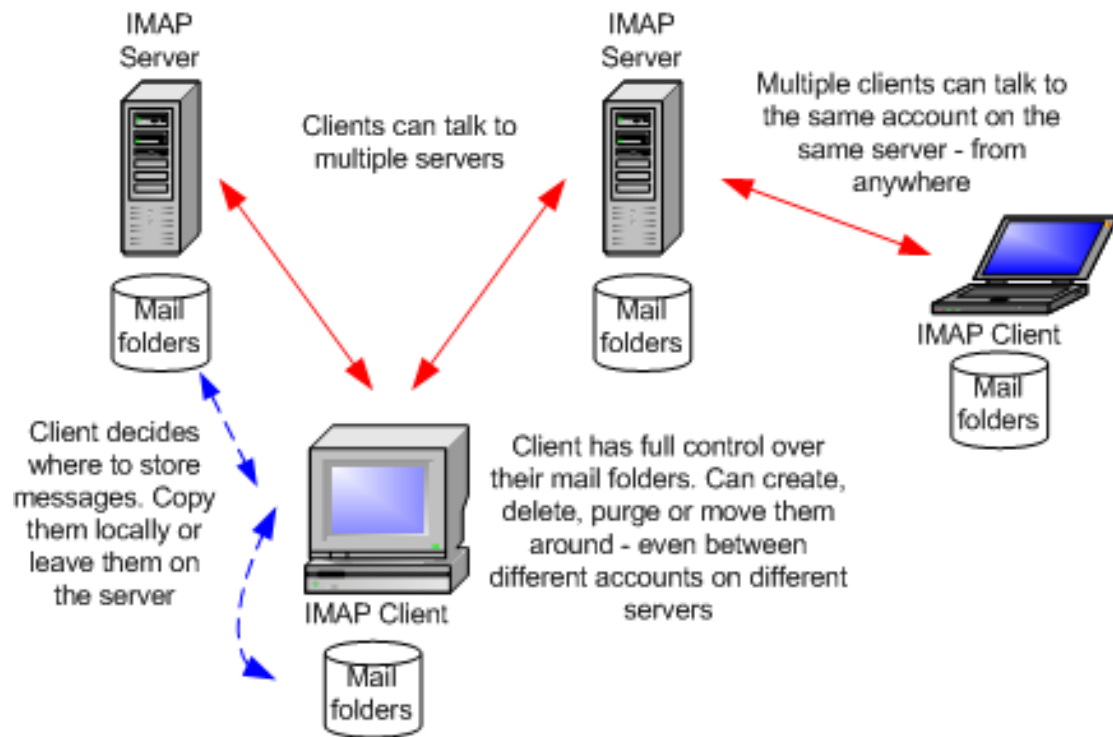
- Barry Leiba



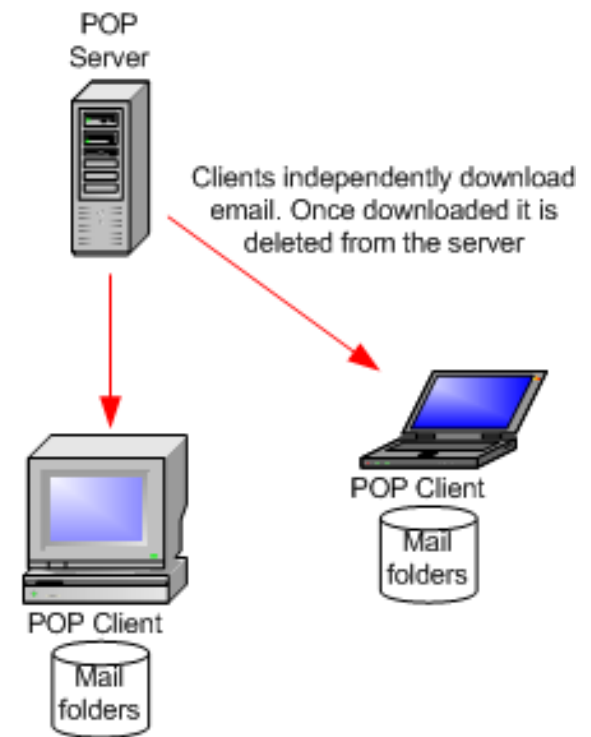
- runs over TCP
- Well-known ports
 - 143: STARTTLS
 - 993: TLS
- Authentication
 - Normal password
 - Encrypted password
 - Kerberos
 - NTLM
 - TLS Certificate
 - OAuth2

- IMAP clients can create, rename, and/or delete mailboxes on the server, and copy messages between mailboxes
 - Mail in Sent folder is pushed from UA towards server!
- IMAP clients often stay connected as long as the user interface is active and download message content on demand
- POP needs exclusive access, while IMAP supports simultaneous sessions to the same inbox
- IMAP protocol allows clients to retrieve any of the individual MIME parts separately and also to retrieve portions of either individual parts or the entire message
- IMAP provides a mechanism for a client to ask the server to search for messages meeting a variety of criteria

IMAP



POP



Command	Description
LOGIN	Authenticates a user to the server
SELECT	Selects a mailbox for read/write access
EXAMINE	Selects a mailbox in read-only mode
CREATE	Creates a new mailbox
DELETE	Deletes an existing mailbox
RENAME	Renames a mailbox
SUBSCRIBE	Marks a mailbox as one to be listed by LSUB
UNSUBSCRIBE	Removes a mailbox from the subscription list
LIST	Lists available mailboxes
LSUB	Lists subscribed mailboxes
STATUS	Retrieves mailbox status information
APPEND	Adds a message to a mailbox
CHECK	Asks the server to perform a checkpoint of the currently selected mailbox
CLOSE	Closes the currently selected mailbox, expunging messages marked for deletion
EXPUNGE	Permanently removes messages marked for deletion
SEARCH	Searches for messages that match given criteria
FETCH	Retrieves data associated with a message, such as headers or flags
STORE	Alters data associated with a message (e.g., setting flags)
COPY	Copies messages to another mailbox
UID	Executes commands using unique identifiers for messages
NOOP	Performs no operation; used to keep the connection alive
LOGOUT	Terminates the session

Response	Category	Description
* OK	Untagged	Indicates that the server is ready or that the previous command completed successfully.
* PREAUTH	Untagged	Indicates that the server has pre-authenticated the connection, allowing the session to start without a separate login.
* BYE	Untagged	Indicates that the server is closing the connection, usually due to an error or shutdown.
OK	Tagged	Confirms that a client-issued command completed successfully.
NO	Tagged	Indicates that the command was understood but could not be completed, typically due to a temporary failure or condition.
BAD	Tagged	Indicates a protocol or syntax error; the server did not understand the command.
+	Continuation	Signals that the server is ready to accept additional data (often used when the server awaits more information from the client).

MAIL PROBLEMS

- **Email Spoofing**
 - Attackers can easily forge the "From" address in an email, making it appear as though the message comes from a trusted sender. This undermines the trust users have in email communications.
- **Forwarding and Relay Issues**
 - Emails that are forwarded or relayed through multiple servers can lose or alter authentication data, making it difficult to verify the original sender's identity.
- **Compromised Accounts**
 - Even when authentication protocols are in place, a legitimate account may be compromised. This allows attackers to send genuine-looking emails from a valid domain, bypassing many checks.
- **Complexity and Misconfigurations**
 - Setting up and maintaining email authentication systems is complex. Misconfigurations (such as incorrect DNS records) can lead to false positives or negatives, causing legitimate emails to be marked as suspicious or allowing illegitimate emails to pass as authentic.

- **Unencrypted Transmission**

- Emails are often sent over unencrypted channels, especially when legacy systems are in use. This makes them vulnerable to interception by unauthorized parties.
- Inadequate or misconfigured encryption settings (for example, not enforcing TLS or using outdated encryption methods) can weaken confidentiality protections and allow data to be compromised.

- **Storage Vulnerabilities**

- Once received, emails are stored on multiple servers—both on the sender's and receiver's side. If these storage systems are not adequately secured, attackers or insiders with access might view confidential content.
- Many email systems do not implement end-to-end encryption by default. Without it, emails can be decrypted at intermediary points like mail servers, increasing the risk of unauthorized access.

- **Third-Party Access**

- Service providers may have policies or vulnerabilities that allow access to stored emails. This means that even if emails are encrypted in transit, they might be decrypted and inspected by third parties once they reach a server.

- **Phishing and Social Engineering**

- Users may inadvertently expose confidential information by responding to phishing emails or through social engineering that trick them into disclosing sensitive data.

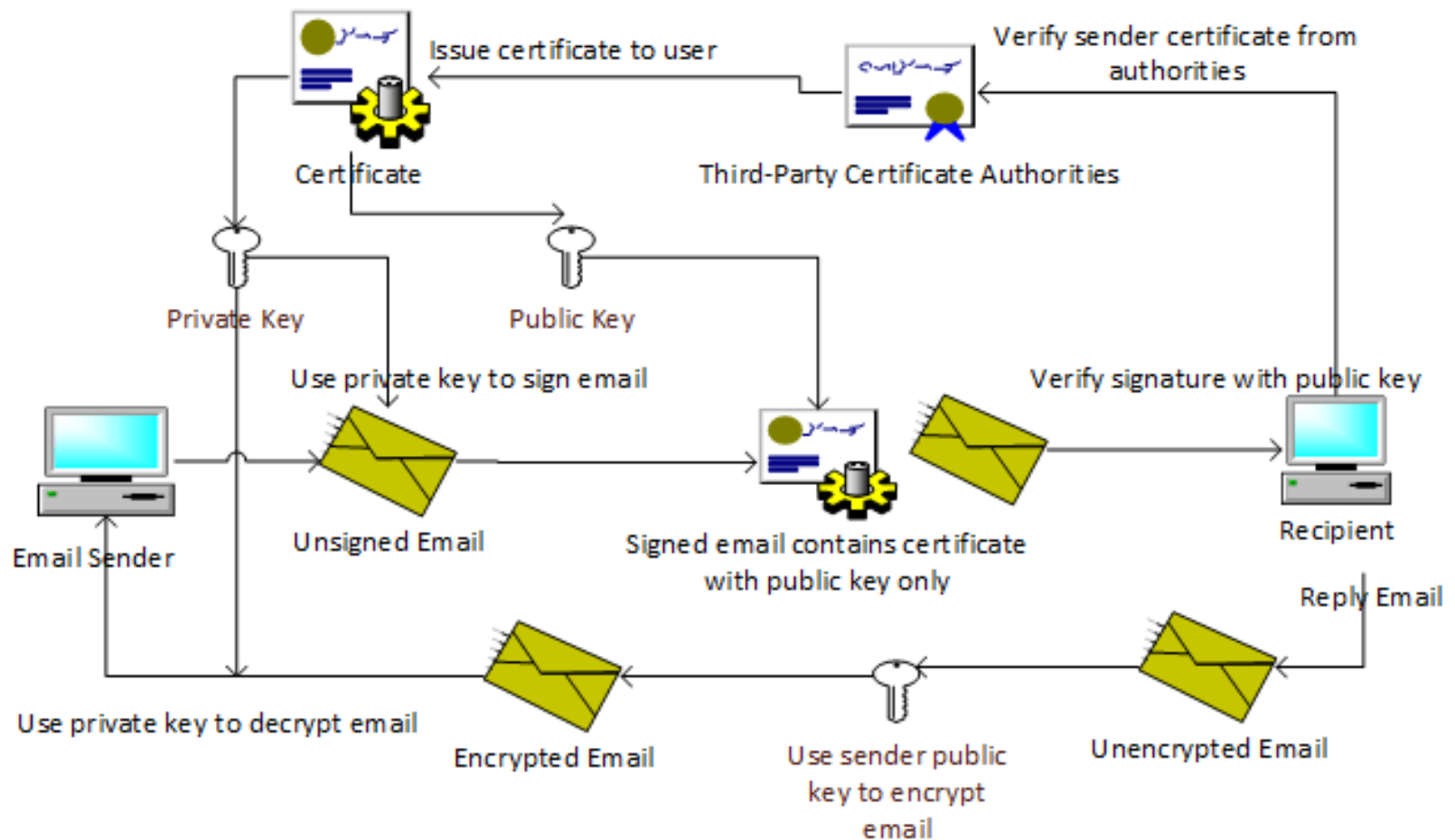
- **Sender Impersonation**
 - Without systems that confirm an email's true origin, malicious senders can easily disguise themselves as legitimate entities. This makes it far easier to trick recipients into trusting fraudulent emails.
- **Unreliable Trust Signals and Diminished Reputation**
 - When there's no reliable way to verify that an email is coming from a genuine source, filtering systems lack the trusted markers they need to effectively distinguish between legitimate messages and spam. This often results in more unwanted messages reaching inboxes.
- **Lack of Accountability**
 - In the absence of mechanisms that provide feedback or reports on authentication failures, domain owners cannot easily detect or respond to abuse. This lack of accountability enables spammers to repeatedly exploit vulnerabilities.
- **Ease of Mass Distribution**
 - Without checks to validate who is allowed to send messages on behalf of a domain, attackers can use compromised systems or botnets to distribute large volumes of spam, overwhelming both mail servers and end users.

- In 2025, spam accounted for roughly 52% of all emails sent worldwide, up from 45.6% in 2023 — meaning more than half of global email traffic is now unwanted (with total daily volume having grown to 376.4 billion emails)
 - The majority of people (96.8%) have received spam messages in some form
- Russia now sends the most spam emails, generating 32.5% of the global total, followed by China (19.1%) and the United States (10.57%) — this is a notable reordering from the original US/China ranking
- The most common topic of spam emails is still prizes and giveaways, followed by job opportunities, then banking
- Over two-thirds (68.8%) of people who received spam/phishing messages reported their mental health being impacted at least a little
- Professional services are now the most targeted sector for business phishing attacks (18%), followed by manufacturing (16%) and retail/wholesale (9%) — financial institutions have dropped out of the top three, now sitting around 6%
 - Delivery services remain the most common subject of spam text messages in the U.S., with over 1.1 billion scam texts on this topic
- Men aged 65+ still receive the most spam calls (35.5/month on average), while men 18-34 receive the fewest (14.2)
- The ten countries sending the most spam emails still produce 2,184 metric tonnes of CO2 daily (797,160 annually) — equivalent to 5.3 million miles driven daily, or 1.9 billion miles a year
- <https://www.emailtooltester.com/en/blog/spam-statistics/>

End-to-End Security

SMIME AND PGP

- RFC 8551 for **securing email through encryption** (achieving confidentiality) **and/or digital signatures** (achieving authenticity).
- **Relies on Public Key Infrastructure** (PKI) to manage digital certificates and cryptographic keys.
 - The sender encrypts the email using the recipient's public key and attaches a digital signature created with the sender's private key.
 - The recipient uses their private key to decrypt the message and the sender's public certificate to verify the signature.
- Widely supported by major email clients, ensuring broad compatibility



```
1 From: =?iso-8859-2?Q?Vladim=EDr_Vesel=FD?= <ivesely@fit.vutbr.cz>
2 To: <igreg@fit.vutbr.cz>
3 Subject: Podepisuji
4 Date: Thu, 21 Apr 2016 11:40:14 +0200
5 Message-ID: <00c101d19bb1$cf051af0$6d0f50d0$@fit.vutbr.cz>
6 X-Mailer: Microsoft Outlook 15.0
7 Thread-Index: AQFBEbyf1hG/EXV4TsYPHwoYB2kvBA==
8 X-OldEid: 0000000A3E822C1B3A5454DA5BB6F5AE6D8DB170700C3B68E10F77511CEB4CD00A
9 MIME-Version: 1.0
10 Content-Language: cs
11 Content-Type: multipart/signed;
12     protocol="application/x-pkcs7-signature";
13     micalg=SHA1;
14     boundary="-----_NextPart_000_00B5_01D19BC2.8F3BB2B0"
15
16 This is a multipart message in MIME format.
17
18 -----_NextPart_000_00B5_01D19BC2.8F3BB2B0
19 Content-Type: multipart/alternative;
20     boundary="-----_NextPart_001_00B6_01D19BC2.8F3BB2B0"
21
22 -----_NextPart_001_00B6_01D19BC2.8F3BB2B0
23 Content-Type: text/plain;
24     charset="iso-8859-2"
25 Content-Transfer-Encoding: 7bit
26
27 Ahoj
28
29 -----_NextPart_000_00B5_01D19BC2.8F3BB2B0
30 Content-Type: application/pkcs7-signature;
31     name="smime.p7s"
32 Content-Transfer-Encoding: base64
33 Content-Disposition: attachment;
34     filename="smime.p7s"
35
36
37 MIAGCSqGSib3DQEHAqCAMIACAQExCzAJBgUrDgMCGGUAMIAGCSqGSib3DQEHAQAoIIOTTCCBuYw
38 ggTOoAMCAQICBF7c4DswDQYJKoZIhvcNAQELBQAwgaoxCzAJBgNVBAYTAKNaMQ0wCwYDVQQHDARC
39 cm5vMRcwFQYDVQQIDA5DemVjaCBSZXh1YmtpYzEmMCQGA1UECgwdQnJubyBVbm12ZXJzaXR5IG9m
```

- PGP created by Phil Zimmermann, who became a subject of U.S.Gov investigation for allegedly violating the *Arms Export Control Act*
- RFC 9580 and OpenPGP provides encryption with public key or symmetric cryptographic algorithms, digital signatures, compression, and key management.



Function	Algorithms Used	Description
Digital signature	DSS/SHA or RSA/SHA	A hash code of a message is created using SHA-1. This message digest is encrypted using DSS or RSA with the sender's private key, and included with the message.
Message encryption	CAST or IDEA or Three-key Triple DES with Diffie-Hellman or RSA	A message is encrypted using CAST-128 or IDEA or 3DES with a one-time session key generated by the sender. The session key is encrypted using Diffie-Hellman or RSA with the recipient's public key, and included with the message.
Compression	ZIP	A message may be compressed, for storage or transmission, using ZIP.
Email compatibility	Radix 64 conversion	To provide transparency for email applications, an encrypted message may be converted to an ASCII string using radix 64 conversion.
Segmentation	—	To accommodate maximum message size limitations, PGP performs segmentation and reassembly.

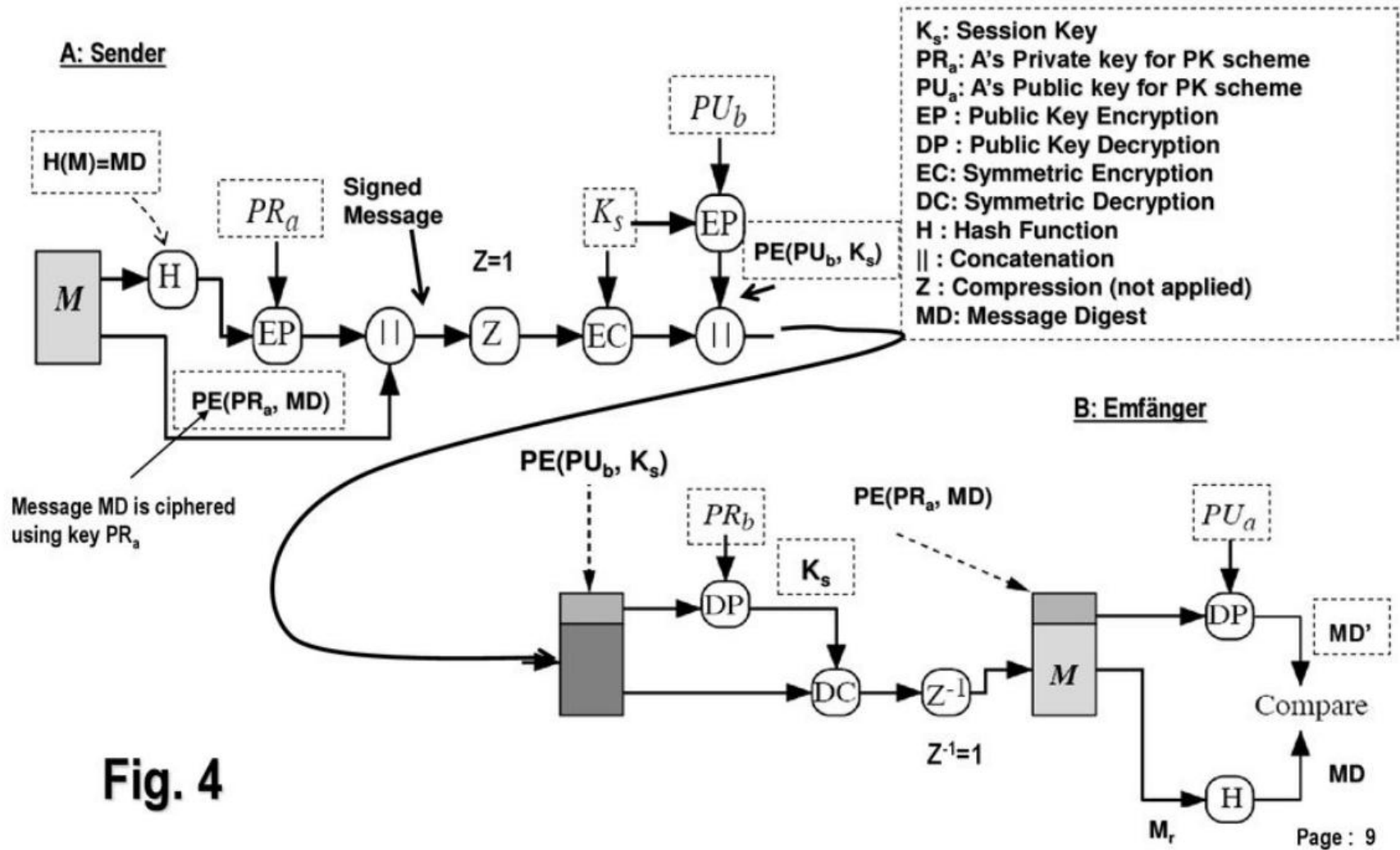
- As a default, PGP compresses the message, using the ZIP compression algorithm, after applying the signature and before encryption.



- PGP uses radix-64 conversion to map groups of three octets into four ASCII characters. Also appends a CRC for data error checking.
- PGP automatically segments messages that are too large. This is done after all of the other steps, including radix-64 conversion. Thus, signature and session key appear only once.

- **Key Management**

- A crucial aspect of PGP is its decentralized approach to key management, PGP does not leverages PKI
 - Users generate their own key pairs and share public keys through key servers or directly.
 - Trust is established through a "web of trust," where users endorse each other's public keys rather than relying on a centralized authority.
 - <https://pgp.mit.edu/pks/lookup?search=polcak&op=index>
- While PGP is most commonly associated with email encryption, it is also used for securing files, communications, and data storage.



```
From: =?UTF-8?Q?Vladim=C3=ADr_Vesel=C3=BD?= <veselyv@fit.vutbr.cz>
To: <hakim.boulahya@spaceapplications.com>
Subject: PGP Test
Date: Thu, 20 Feb 2020 09:25:04 +0100
Message-ID: <00b301d5e7c7$41787860$c4696920$@fit.vutbr.cz>
MIME-Version: 1.0
Content-Type: multipart/alternative;
    boundary="-----_NextPart_000_00B4_01D5E7CF.A33DA3B0"
X-Mailer: Microsoft Outlook 16.0
Content-Language: cs
Thread-Index: AQJW23DQvV+fS5yR9G/oAUo4DDPCbA==
X-OlkEid: 00000000282BD258CC379843A5AA4B01824A779E0700C3B68E10F7751
```

This is a multipart message in MIME format.

```
-----_NextPart_000_00B4_01D5E7CF.A33DA3B0
Content-Type: text/plain;
    charset="UTF-8"
Content-Transfer-Encoding: 7bit
```

-----BEGIN PGP MESSAGE-----

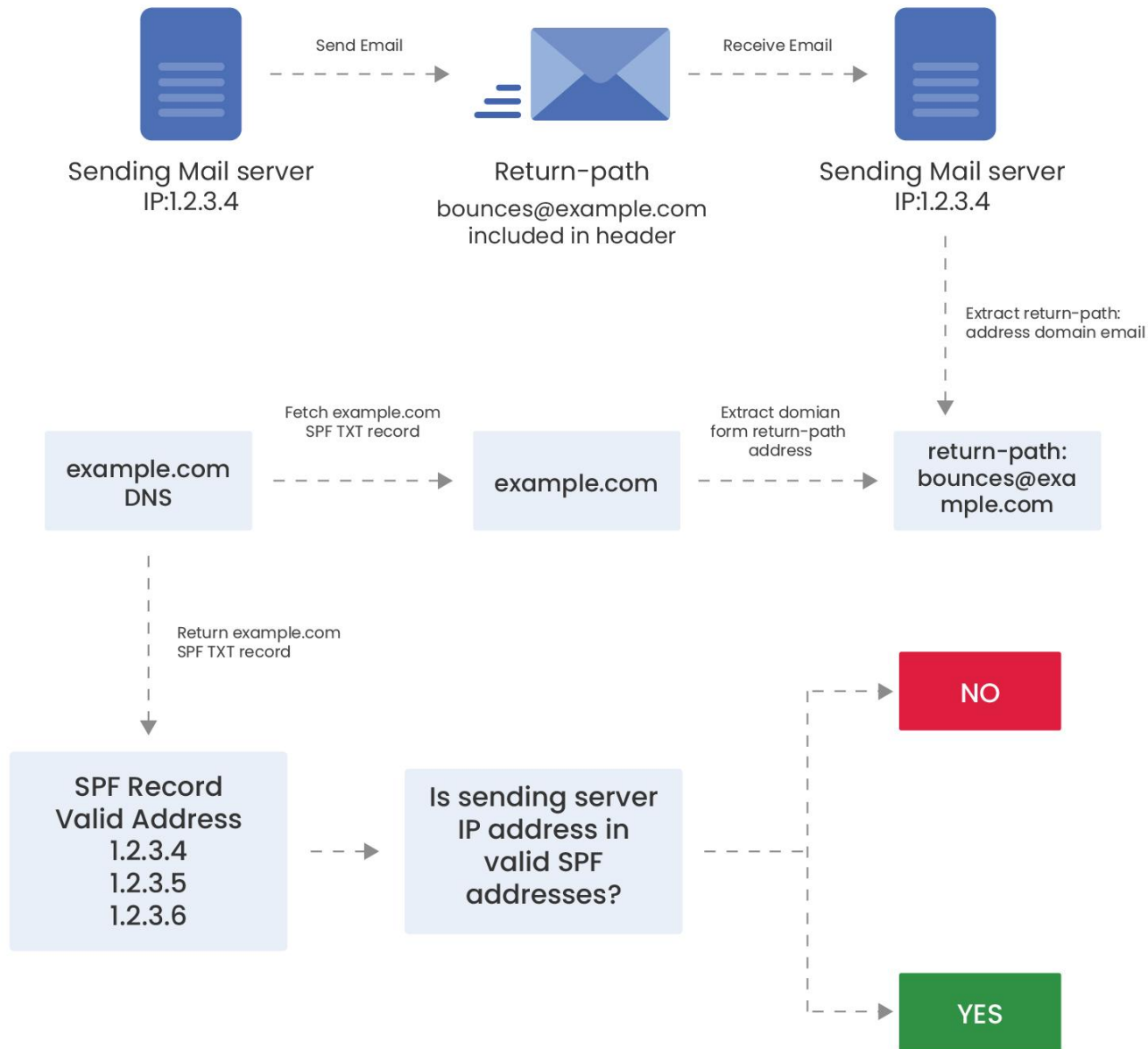
```
hQIMA8RFUfwuEczXARAAyW9Xq7mZ3fJo+lk2/ULv9opT+hRjXBJL0Uq+fQ1rs6nn
AaNiV/HT/TP0BatmQKE8mgcxwh23VkDkUjo1UxmxcJdKen5LUIbMuhlz4PGXk5sC
+JLA4wPlXQwlen/Fg6Us5BGKH6Z5Z9RRRQA1xnj7FZaRfZP8vSdTARcO3AbnOvYA
kkyPTjkuzyJIr5CqLhC3kMRu1ycCdISsWBceahovcQjSh11gZzJbyHlN97HHidC8
0UsYf2BgnOT6D6s2UmOQF0UuRfoxFbkbm31/RV6DJDFp8NiDvj2IYk6a/bQH5Opw
SLEnAtLbnm/qGnsBddz+mrV+PzBO+y8ORczLywrTS3oCMUHgi5RmJ2zNK/Q4NluE
HxyeTTEwXankum2Wt5vf2prPomlzaVklZdDhy1huF1xteRDWDKA3nDQHR+XI0eLs
```

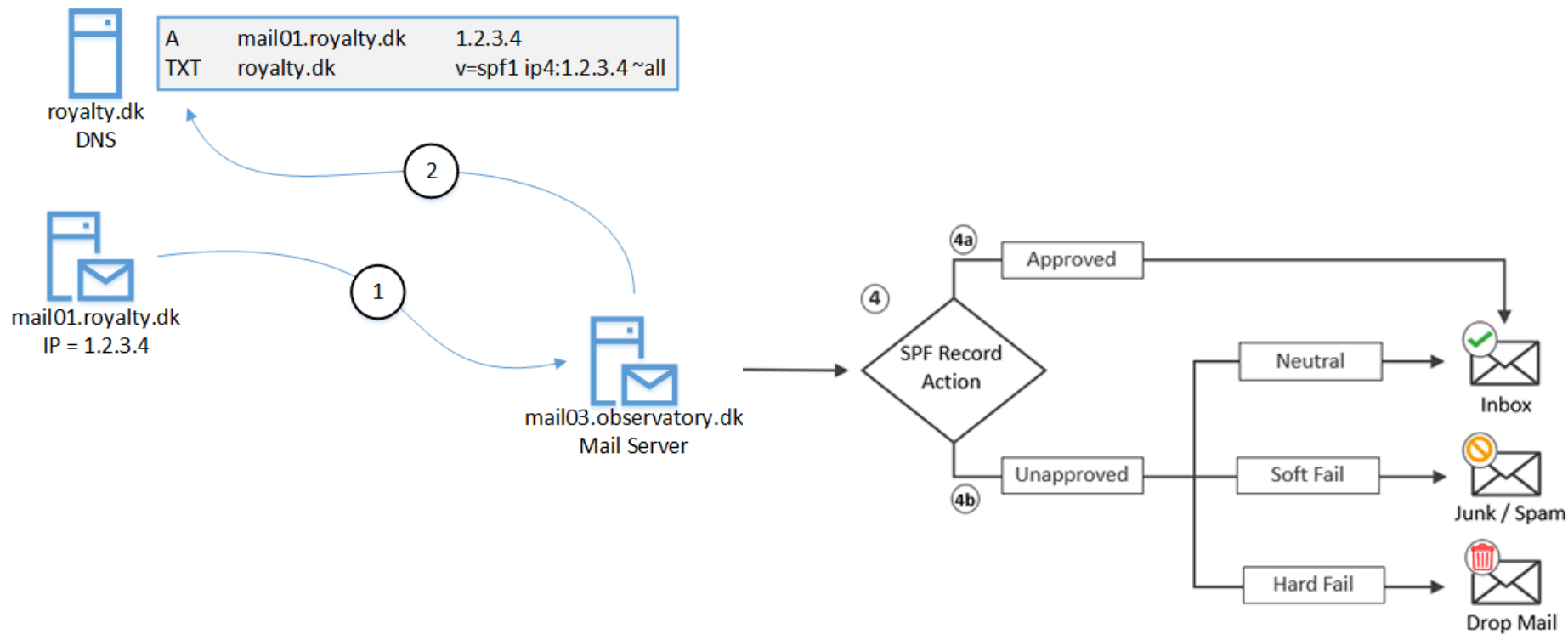
SPF

- **SPF** is an email authentication protocol designed to detect and prevent sender address forgery defined in RFC 7208.
 - It helps email receivers determine if incoming messages from a domain are sent by servers authorized by that domain's administrators.
 - Prevents spoofing and enhances trust
- Improves reputation
 - having SPF is better than not having it
- Enhances security
 - makes it more difficult for attackers to impersonate a domain
- Provides transparency
 - domain owners can monitor unauthorized use by analyzing email logs and SPF validation results

- A domain owner publishes an SPF record in the DNS. This record is stored as a **TXT record** and specifies which IP addresses or hostnames are allowed to send email on behalf of the domain.
- **fit.vutbr.cz**
 - `v=spf1`
`a:kazi.fit.vutbr.cz`
`a:eva.fit.vutbr.cz`
`ip4:147.229.12.8`
`ip6:2001:67c:1220:808::93e5:80c`
`ip4:147.229.176.14`
`ip6:2001:67c:1220:8b0::93e5:b00e`
`a:list.fit.vutbr.cz`
`ip4:147.229.9.34`
`ip6:2001:67c:1220:809::93e5:922 ~all`
- **demo**
 - `v=spf1 ip4:192.0.2.1`
`ip4:198.51.100.0/24 a mx`
`include:mailservice.example.com -all`
- **Syntax and semantics**
 - `v=spf1`: Identifies the version of SPF being used.
 - `ip4:192.0.2.1`: Authorizes a specific IPv4 address.
 - `ip4:198.51.100.0/24`: Authorizes a range of IPv4 addresses.
 - `a`: Authorizes the IP address found in the A record of the domain.
 - `mx`: Permits servers listed in the domain's MX records.
 - `include`: Authorizes external domains (such as third-party email services) to send on behalf of the domain.
 - `-all`: Indicates that any server not matched by the earlier mechanisms is not permitted to send email for this domain (a hard fail).

- Mechanisms define the methods to authorize senders
 - `ip4/ip6`: Specify individual IP addresses or ranges.
 - `a`: Authorizes the IP address returned by the domain's A record.
 - `mx`: Uses the IP addresses in the domain's MX records.
 - `include`: Permits other domains' SPF records to be included in the evaluation.
- Qualifiers modify the effect of each mechanism
 - `+`: Pass (this is the default and can be omitted).
 - `-`: Fail; messages from non-authorized IPs should be rejected.
 - `~`: SoftFail; messages might be accepted but marked as suspicious.
 - `?`: Neutral; no definitive judgment is made.



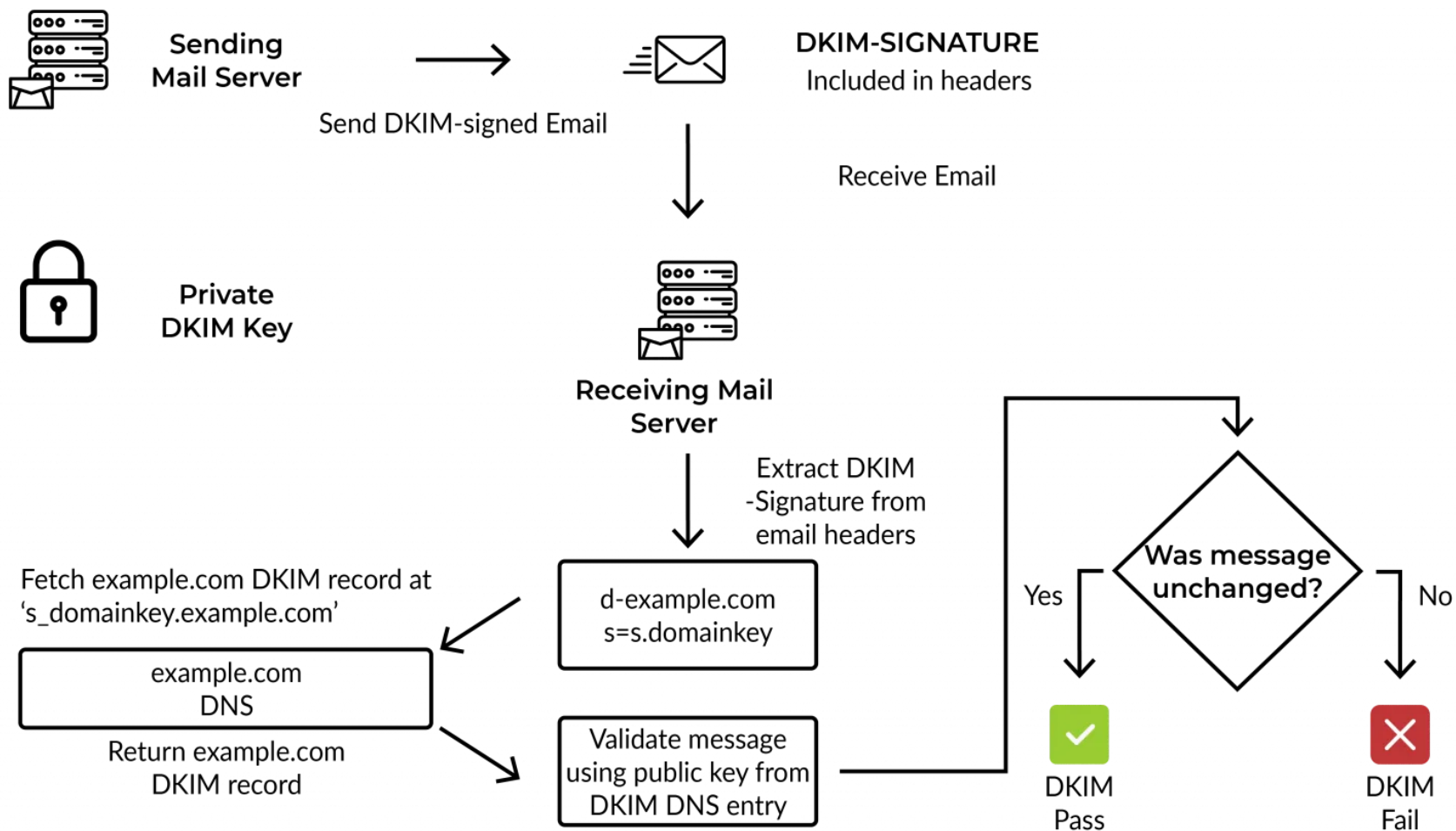


DKIM

- **DKIM** (standardized in RFC 6376) is an email authentication method that uses cryptographic techniques to verify that a message was indeed sent by the proper domain and that its contents have not been altered during transit.
- **Email Integrity**
 - DKIM ensures that an email's content remains unchanged from the moment it is signed by the sender to when it is received. This provides assurance that the message has not been tampered with.
- **Sender Authentication**
 - By verifying the digital signature in the email header, recipients can confirm that the email originates from a domain authorized by the sender, reducing the likelihood of spoofed messages.
- **Enhancing Trust and Deliverability**
 - When mail servers see that an email passes DKIM checks, they are more likely to consider it trustworthy, which can help improve deliverability and reduce spam filtering.

- Private/Public key pair RSA/ED25519 for signing, SHA-1/SHA-256 for integrity
- Special DNS DKIM record of type TXT
 - DKIM records are stored under a specialized name
 - `[selector]_domainkey.[domain]`
 - Record value consist of:
 - v=DKIM1: Version of DKIM being used, always set to "1"
 - k: Key type
 - p: The public key used for verification, represented as a long string of characters
 - <https://mxtoolbox.com/SuperTool.aspx?action=dkim%3afit.vutbr.cz%3amx1&run=toolpage>
- Mail server
 - Upload private key, which is used for signing

- The digital signature is attached to the email as a new header field
- This header includes several parameters
 - v=DKIM1: Specifies the version.
 - a=: Indicates the signing algorithm (often RSA with SHA-256).
 - d: The signing domain.
 - s: The selector that identifies which public key to use from the DNS.
 - h: Lists the header fields included in the signature.
 - b: Contains the actual digital signature (encoded).
 - c: Defines the canonicalization algorithms applied to the header and body, where the first part is for the header and the second for the body.
 - t: A numeric value representing the time the signature was generated.
 - x: Specifies the time after which the signature should no longer be considered valid.
 - l: Indicates the length of the canonicalized body that was used to compute the signature.
 - q: Defines the method used to retrieve the public key. The default is usually `dns/txt`.
 - bh: Contains the hash of the canonicalized body, which is used to verify that the body hasn't been altered.
 - i: Specifies the identity on behalf of which the message is signed, usually in the form of an email address



• Failed

```
Return-Path: <fitinfo-bounces@list.fit.vut.cz>
Received: from list.fit.vutbr.cz (list.fit.vutbr.cz
[IPv6:2001:67c:1220:809:0:0:93e5:922])
    by kazi.fit.vutbr.cz (8.18.1/8.18.1) with ESMTPS id 51SDLIDc083916
    (version=TLSv1.3 cipher=TLS_AES_256_GCM_SHA384 bits=256 verify=NO);
    Fri, 28 Feb 2025 14:21:18 +0100 (CET)
    (envelope-from fitinfo-bounces@list.fit.vut.cz)
DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed; d=fit.vut.cz; s=mx1;
t=1740748878; bh=K1p4MAijJ6hVMethbtkKCN53z9/Y20j9zRfW0963V2g=;
h=Date:From:To:Subject:List-Id:List-Archive:List-Help:List-Owner:
List-Post:List-Subscribe:List-Unsubscribe;
b=QjLLJoMX1tu9btGXlbgA85456Tm8W53e1ARmz+CW0DBZp78bVY5nvQD6ucdSC12xw
Tf4ri2A0jQ/KHZ1cuq6TbQDglV+LuAo/vioqE6WnpHJQCCHHRcxCMpBoF+AKbHGr/fk
L9l/gjb0xYGCuTP5/oniYapSrtM0g8ZZtlnVw16Q=
Authentication-Results: list.fit.vutbr.cz;
    dkim=fail reason="signature verification failed" (1024-bit key; secure)
    header.d=fit.vut.cz header.i=@fit.vut.cz header.a=rsa-sha256
    header.s=mx1 header.b=utxXjQ3r; dkim-atps=neutral
```

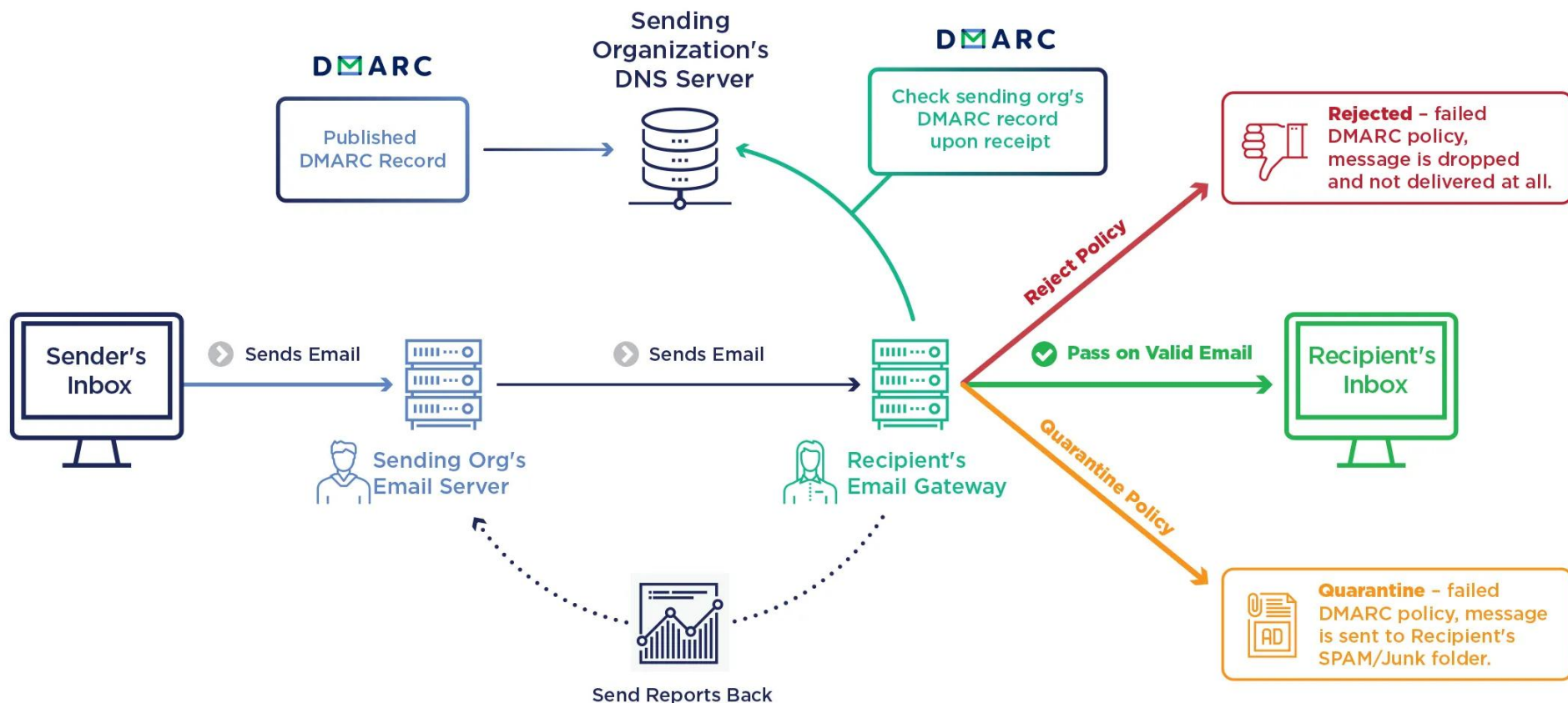
• Passed

```
Return-Path: <burgetr@fit.vut.cz>
Received: from pcburgetr.localnet (
[IPv6:2001:67c:1220:80c:dc2:bde3:f070:2f0c])
    (authenticated bits=0)
    by kazi.fit.vutbr.cz (8.18.1/8.18.1) with ESMTPSA id 51IDm5bs052616
    (version=TLSv1.3 cipher=TLS_AES_256_GCM_SHA384 bits=256 verify=NO);
    Tue, 18 Feb 2025 14:48:05 +0100 (CET)
    (envelope-from burgetr@fit.vut.cz)
DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed; d=fit.vut.cz; s=mx1;
t=1739886485; bh=k+3hwkBDxIgk/6V/K+GpnTb1SoBoE694bXmCrWfvNUA=;
h=From:To:Cc:Subject:Date;
b=CAJ2z0ae9x/hmbg3bVXyoGCH4UHIjcn7WpVTRo6/3BtJ/yptPnPwYs3nk8f2qQRBg
sS9+AoBZNiA+eg2tzA6ZvyLBSCdK3vBSdabALXFAKK5tH0If51HEAzJ+UZ7ubeLn20
Oh//thiGURXs3pN7xbdrsG6nScG5gnSaIBW//5zI=
```


DMARC

- DMARC is defined in RFC 7489 is an email authentication protocol that builds on SPF and DKIM
- Authentication
 - DMARC verifies the authenticity of email senders by aligning the results of SPF and DKIM checks with the domain in the "From" header
- Policy enforcement
 - Domain owners can specify how receiving email servers should handle messages that fail authentication
- Reporting
 - DMARC provides a feedback mechanism, allowing domain owners to receive reports on email authentication results and potential abuse of their domain

- DMARC records are published as DNS TXT records
- They use the following parameters:
 - `v`: Version identifier
 - `p`: Policy for handling unauthenticated emails (`none`, `quarantine`, `reject`)
 - `rua`: Aggregate reporting email address
 - `sp`: Subdomain policy
 - `ruf`: Forensic reporting email address (optional)
 - `fo`: Failure reporting options
 - `pct`: Percentage of messages subject to policy
 - `adkim`: DKIM alignment mode (`strict` or `relaxed`)
 - `aspf`: SPF alignment mode (`strict` or `relaxed`)
- <https://mxtoolbox.com/SuperTool.aspx?action=dmarc%3afit.vutbr.cz&run=toolpage>



- `report_metadata`: Contains information about the reporting organization, contact details, a unique report identifier, and the time period the report covers (expressed in Unix time).
- `policy_published`: Shows the DMARC policy published by the domain owner, including:
 - `domain`: The domain under analysis.
 - `adkim/aspf`: Alignment modes for DKIM and SPF (e.g., "r" for relaxed).
 - `p/sp`: Somain/Subdomain policy applied (e.g., reject).
 - `pct`: The percentage of messages to which the policy is applied.
- `record`: Represents a single set of authentication results:
 - `row`: Contains the IP address of the sender, the number of messages from that IP, and the outcome of policy evaluation (disposition, and whether DKIM and SPF passed or failed).
 - `identifiers`: Specifies the domains found in the envelope (SMTP) and header (mail).
 - `auth_results`: Provides detailed results for DKIM and SPF evaluations for that record.

```
<?xml version="1.0" encoding="UTF-8" ?>
<feedback>
  <report_metadata>
    <org_name>example.com</org_name>
    <email>noreply-dmarc-support@example.com</email>
    <extra_contact_info>http://example.com/dmarc/support</extra_contact_info>
    <report_id>9391651994964116463</report_id>
    <date_range>
      <begin>1335571200</begin>
      <end>1335657599</end>
    </date_range>
  </report_metadata>
  <policy_published>
    <domain>bix-business.com</domain>
    <adkim>r</adkim>
    <aspf>r</aspf>
    <p>none</p>
    <sp>none</sp>
    <pct>100</pct>
  </policy_published>
  <record>
    <row>
      <source_ip>203.0.113.209</source_ip>
      <count>2</count>
      <policy_evaluated>
        <disposition>none</disposition>
        <dkim>fail</dkim>
        <spf>pass</spf>
      </policy_evaluated>
    </row>
    <identifiers>
      <header_from>bix-business.com</header_from>
    </identifiers>
    <auth_results>
      <dkim>
        <domain>bix-business.com</domain>
        <result>fail</result>
        <human_result></human_result>
      </dkim>
      <spf>
        <domain>bix-business.com</domain>
        <result>pass</result>
      </spf>
    </auth_results>
  </record>
</feedback>
```

Q & A

- Describe DNS messages format
- Describe at least 3 DNS records
- What is DNSSEC and what it offers?
- What is DoH and what it offers?
- What is mail user-agent and what features it has? Name examples.
- Describe email structure. What is MIME?
- Describe steps in sending mail from one user and receiving it by another user.
- Describe in detail SMTP protocol.
- What is IMAP? What is POP? How they differ?
- What is S/MIME? What is PGP? How they differ?
- Describe generation and obtaining of PGP key.
- What is SPF and how it protect email exchange?
- What is DKIM and how it protect email exchange?
- What is DMARC and how it protect email exchange?

- DNS
 - [RFC 1034](#), [RFC 1035](#), [RFC 2915](#), [RFC 3403](#), [RFC 3236](#), [RFC 2782](#), [RFC 4408](#), [RFC 3596](#), [RFC 3833](#)
 - DNSSEC
 - [RFC 4033](#), [RFC 4034](#), [RFC 4035](#), [RFC 5155](#)
 - DNS over TLS
 - [RFC 7858](#), [RFC 8310](#)
 - DNS over HTTPS
 - [RFC 8484](#)
 - DNS over QUIC
 - [RFC 9250](#)
 - [DoH story thesis](#)
- <https://www.eecis.udel.edu/~amer/856/smtp.04f.ppt>
- <https://www.geeksforgeeks.org/simple-mail-transfer-protocol-smtp/>
- <https://threatcop.com/blog/spf-authentication/>
- <https://emailauth.io/dkim>
- RHOTON, John. *Programmer's guide to internet mail: SMTP, POP, IMAP, and LDAP*. Digital Press, 1999.

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- <https://www.fit.vut.cz/research/groups/nes@fit/>
- <https://git.fit.vutbr.cz/NESFIT>
- <https://nesfit.github.io>